Melati

Multilingual **E**xpressive speech, **L**abelled for **A**ffect, **T**ags and **I**nterjections

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ABSTRACT

Pipelines that turn podcasts into speech-synthesis data, Emilia-Pipe first among them, keep the words and lose what happens between them: laughs and fillers are separated away or never written, emotion is not labelled, and the transcript of a pause is whatever Whisper invents. We present Melati, a pipeline and a dataset of Malaysian podcast speech whose transcripts carry timed tags for non-verbal sounds, emotion spans and the fillers as spoken. Every model in it is a server. On VoxConverse, against the same recordings run through upstream Emilia-Pipe, it keeps 87% of the reference single-speaker speech against 22%, puts a second speaker in 2.2% of its segments against 19.5%, and starts mid-word in 9.2% of them against 37.1%. Its transcriber, released on Hugging Face, is a merge of two whisper-large-v3 fine-tunes that writes text over 5.1% of real voice-free audio against 99.9% for base, with lower error on Malaysian speech (21.6% against 23.4% WER) and on FLEURS. Measuring it as vLLM serves it exposed that Hugging Face's default decoding forbids an empty transcript, which hid both the fix and its cost: without replay of other languages, a fine-tune treats an unfamiliar language as non-speech and returns nothing for 29.6% of FLEURS. Emotion is tagged by seven models averaged, five of them heads we trained, reaching 0.597 unweighted recall on held-out corpora against 0.529 for the released classifiers. Breaths come from a classifier we trained on word-gap sounds, laughs and coughs from a frame-level head that hears them over speech too: on held-out podcasts, the pipeline finds 59.6% of the laughs and 70.1% of the breaths (87% and 83% precise where labelled); Qwen3-Omni finds 46.8% of the laughs and no breaths. Every tag is kept only where its precision was measured. The source code of Melati is at https://github.com/Scicom-AI-Enterprise-Organization/Melati.

1 Introduction

A speech-synthesis model can only say what its training transcripts write down. Emilia [He et al., 2024, 2025] showed how to turn in-the-wild speech into such transcripts at scale, and its pipeline, Emilia-Pipe, has been run on Malaysian podcasts before (Malaysian-Emilia v1 and v2). Three things are missing from what it produces: the sounds between words (a laugh, a breath, a cough), the fillers ("erm", "emm"), and the emotion of a phrase. Modern speech generators read all three as inline tags [Han et al., 2026]. Figure 1 is the pipeline we built to produce them.

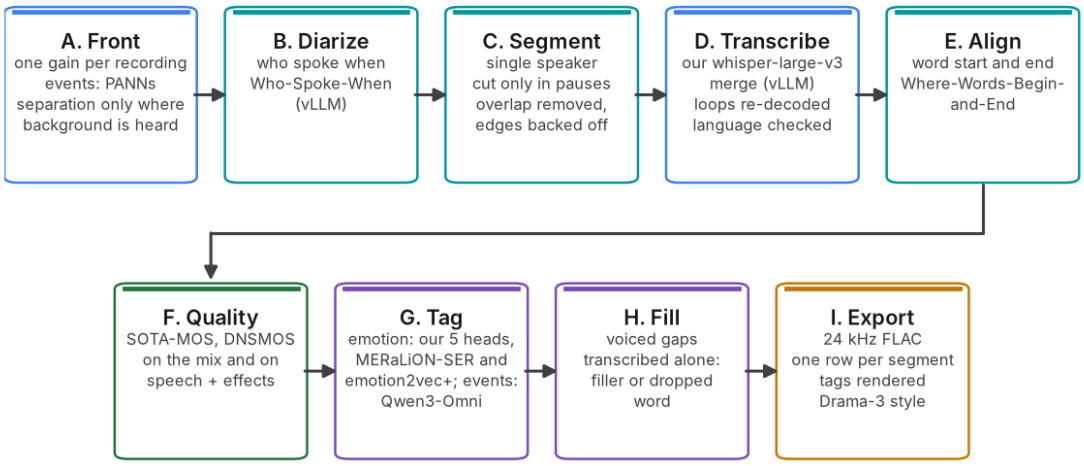


FIGURE 1 Melati-Pipe. Nine stages, each a client of a model server (vLLM for the two Whisper-sized decoders, dynamic-batching servers for the rest). Stage letters are used throughout the paper.

What a row of the dataset looks like, in the tag format of Fish Audio Drama 3 [Han et al., 2026]: point tags in square brackets act at a position, scoped tags in angle brackets act on the words they enclose, fillers stay words.

```
<happy>hello [coughs] sorry, um, I just got the results</happy> [laughs]
```

The tags are a rendering. The record underneath is timed: every word, filler and event has a start and an end, and every emotion span names the words it covers, so another format is a re-rendering, not a re-annotation.

Our contributions:

- An audit of what Emilia-Pipe does to the audio and to the text, measured on its released Malaysian output and on the same recordings run through it (Section 2).
- A pipeline that keeps the expressive content and every model behind an API, with segmentation that cuts only in pauses and backs off from removed overlaps (Section 3).
- A released whisper-large-v3 merge that does not hallucinate over non-speech; three defects in the training recipe it follows; the finding that the default decoding hid both the fix and its cost, a model that stops on languages it never heard; and a merge of two fine-tunes that is better than either (Section 4).
- Emotion heads on five frozen encoders whose average tags five emotions at measured precision where the released classifiers could tag three; a non-verbal classifier trained on the pipeline's own word-gap candidates, a gap finder that reaches the breaths it missed, and a frame-level head for the laughs and coughs over speech, where no gap holds them; filler tags with measured precision (Section 5).
- A released AudioSet clip tagger, distilled from the average of released ones and then of six teachers, two of them its own earlier versions: 0.5152 mAP on the same evaluation clips where the best released one scores 0.4997. The six teachers' average is released too, at 0.5219 (Section 5.4).
- A side-by-side run of both pipelines on the same recordings (Section 6).

2 What Emilia-Pipe does to speech

Emilia-Pipe standardises a recording, separates vocals with UVR-MDX [Kim et al., 2021], diarizes it with pyannote [Bredin, 2023], cuts it with Silero VAD [Silero Team, 2024], transcribes it with Whisper [Radford et al., 2023] and keeps the segments DNSMOS [Reddy et al., 2022] scores at 3.0 or more. We read its code line by line (Table 1) and measured 2,000 segments of each released Malaysian-Emilia version.

TABLE 1 Defects of upstream Emilia-Pipe, and what Melati does instead. "Fork" is the mesolitica fork that produced Malaysian-Emilia v2. Line numbers are in the repository's `docs/emilia_bugs.md`.

upstream	fork	Melati
32 kbit/s MP3 export, low-passed near 8 kHz	fixed	24 kHz FLAC
every segment peak-normalised to full scale	kept	one gain per recording
VAD only on turns over 20 s; hard 30 s cuts	kept	cuts only in pauses, within Whisper's 30 s window
overlapping speech kept in both speakers	kept	overlap removed from both
separation everywhere, vocals stem only	kept	only where background is heard; speech + effects
DNSMOS ≥ 3 applied before export	kept	every score a column, none filtered on
language-ID confidence filters code-switching	kept	re-decode only outside the allowed languages
transcription prompt full of fillers	kept	a Whisper trained not to write over non-speech
one empty recording ends the whole run	fixed	every stage resumes from its own files

The export defect is the easiest to see. Figure 2 is the long-term spectrum of the two releases: v1 stops at 8.4 kHz (its MP3s are 32 kbit/s, Amphion issue #436), v2 reaches 11.3 kHz.

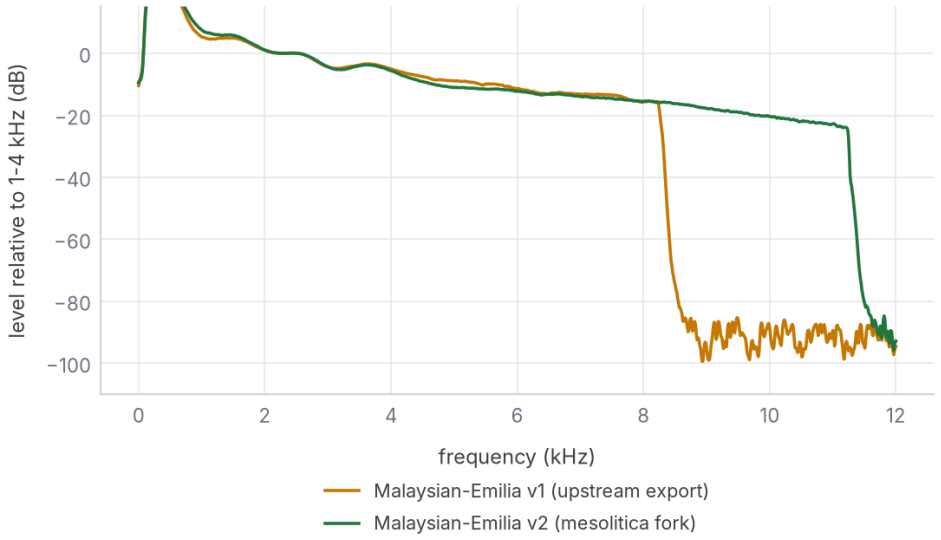


FIGURE 2 Long-term average spectrum of Malaysian-Emilia v1 and v2, relative to its own 1–4 kHz level.

Peak normalisation makes the level of a segment follow its loudest transient, and MP3 then pushes some over full scale. Figure 3 counts the segments that touch it.

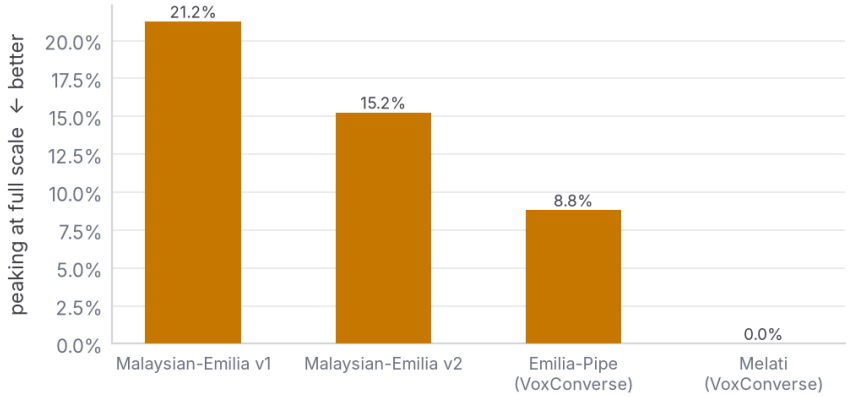


FIGURE 3 Segments whose peak is within 0.1 dB of full scale. The last two bars are the same twelve VoxConverse recordings through each pipeline (Section 6).

3 Melati-Pipe

3.1 EVERY MODEL IS A SERVER

No pipeline code touches a GPU. Whisper and the diarization model run on vLLM [Kwon et al., 2023]; the separator, event detector, aligner, quality and emotion models run behind servers that batch requests as they arrive. A recording walks the stages as a client, and many recordings are in flight at once, so while one is being separated others are being transcribed and every server's batches stay full. Scaling is a replica count or a tensor-parallel size in one configuration file.

3.2 SEPARATION THAT KEEPS THE VOCALISATIONS

Separation exists to remove music. Two things decide where Melati applies it and what it keeps. First, it runs only where PANNs [Kong et al., 2020] hears background (music,

singing, crowd, broadcast noise), because elsewhere the recording itself is cleaner than any separator's output. Second, what it keeps: DBS-TIGER [Li, 2025, Xu et al., 2025], trained on speech, music and effects [Petermann et al., 2022], files a laugh under effects. We placed 403 real vocalisations from FSD50K [Fonseca et al., 2020] in the pause between two utterances of one Common Voice speaker [Ardila et al., 2020] and measured where each separator puts it (Figure 4).

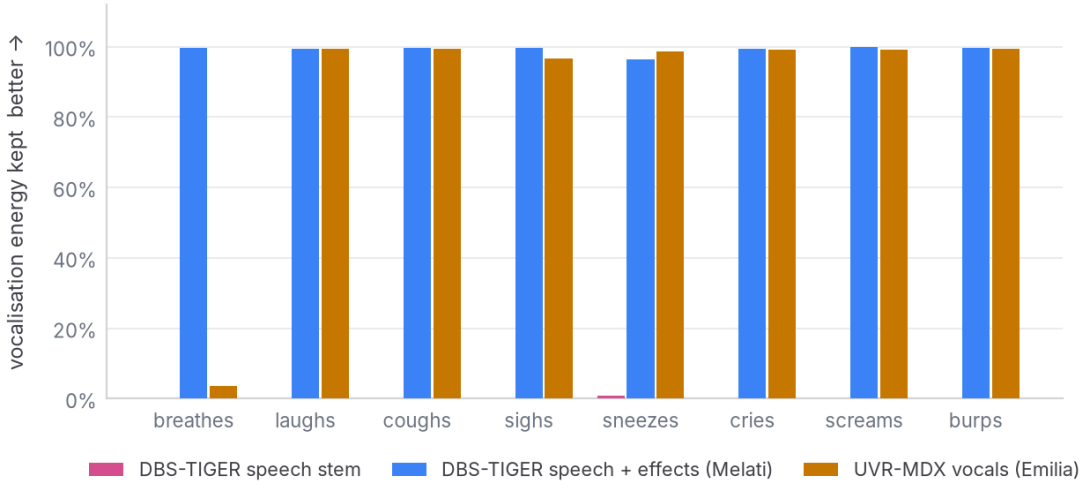


FIGURE 4 Share of an injected vocalisation's energy in the stem each pipeline keeps, no music under it. The speech stem alone keeps at most 0.9%; speech plus effects keeps at least 96%; UVR-MDX keeps every class but breaths (4%).

So Melati's separated version is speech plus effects: only the music goes. Every later stage (quality, the taggers, export) reads that version; transcription, diarization and alignment read the speech stem, where a laugh only gets in the way of the words.

The same stimuli show what a quality filter does to expressive speech. Adding the vocalisation lowers DNSMOS OVRL by 0.04 to 0.53 and takes 15% to 39% of otherwise clean clips under Emilia's 3.0 threshold (Figure 5). A filter applied before export deletes segments for being expressive; Melati keeps every score as a column instead.

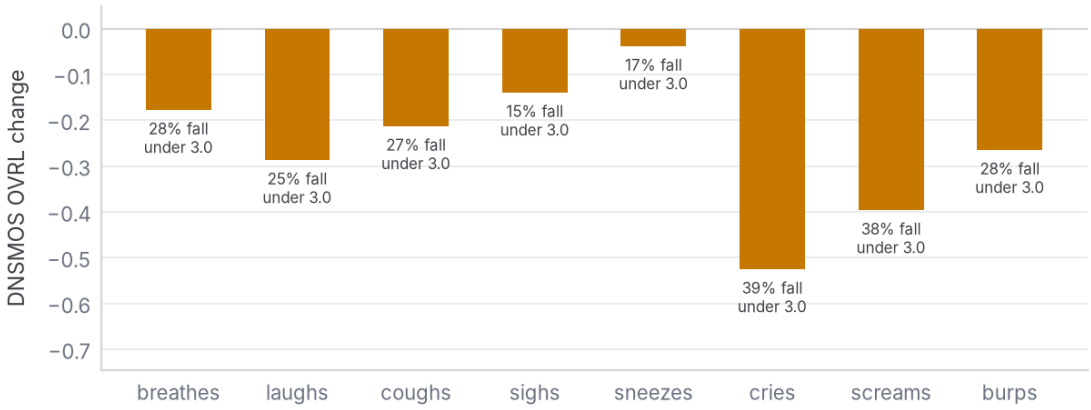


FIGURE 5 Change in DNSMOS OVRL when a vocalisation is added to the pause, and the share of clips it takes from 3.0 or more to under 3.0.

3.3 SPEAKERS AND SEGMENTS

Stage B is Who-Spoke-When [Khaidukova and Zolkepli, 2026], a Whisper-based end-to-end diarizer served by vLLM; a recording up to 45 minutes is one request (80 in the pilot; see Section 7), a longer one is ten-minute windows overlapping by a minute. Stage C turns its turns into segments (Figure 6): a cut falls only in a pause the VAD finds, a segment never exceeds Whisper's 30-second window, and wherever two speakers overlap the overlap is removed from both, so no segment carries a second voice. Removing an overlap cuts the first speaker off mid-word, so that edge moves back to the least speech-like frame within 1.5 s and gets no padding. On the podcast pilot that took the segments ending inside their last word from 18.4% to 15.0%, for 15.26 hours exported instead of 17.70 (same transcriber both times).

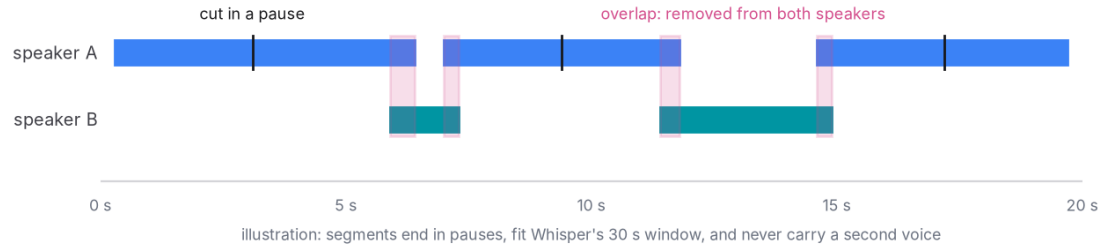


FIGURE 6 Segmentation, as an illustration: cuts in pauses, overlap removed from both speakers.

3.4 WORDS, TIMES AND QUALITY

Stage D is our whisper-large-v3 fine-tune (Section 4). A segment detected outside the allowed languages (Malay, English, Chinese, Tamil) is re-decoded with the allowed language Whisper ranks highest: base large-v3 wrote parliamentary English as Welsh on our development set. On podcasts that re-decode lands on Tamil for Arabic chant, songs and stray Malay, so export keeps Chinese and Tamil only where Whisper heard them itself, in a recording mostly in that language (Section 7). Stage E aligns the transcript with Where-Words-Begin-and-End [Zolkepli, 2026b], which also scores its own confidence. Stage F scores every segment twice, on the mix and on speech plus effects, with SOTA-MOS [Scicom AI Enterprise, 2026] and DNSMOS, and export keeps the mix wherever the music is 25 dB under the voice and the scorer does not prefer the separated version. Export trims a segment to 0.15 s around its first and last sound, never past the pause it was cut in.

4 A Whisper that writes nothing over non-speech

Base whisper-large-v3 writes words over 61.9% of silent clips, 99.0% of music and 99.9% of real voice-free audio. In a pipeline that transcribes every gap between words to find fillers (stage H), that is a filler or a dropped word in every pause. We fine-tune it with the recipe of Zolkepli [2026a]: voice-free audio with empty targets, spoken hallucination phrases with their true text, LoRA [Hu et al., 2022] rank 64 on the attention and MLP projections, 1,000 steps of 16 clips. Its paper reports a multilingual loss (FLEURS CER up for every configuration), so we add Common Voice 22 in the 77 languages it shares with Whisper, replayed in the same batches. The mixes are 58,041 clips (no replay, 46% voice-free) and 77,388 (25% replay).

4.1 THREE DEFECTS IN THE RECIPE

The first is in what the decoder reads. The recipe masks the four prompt tokens out of the loss by setting them to −100 in the labels, and lets Hugging Face build the decoder inputs from the labels. That builder shifts the labels right and replaces every −100 with the pad token. So in training the decoder never sees `<|ms|><|transcribe|><|notimestamps|>`: it sees padding, and the text starts one position later than it will at inference (Figure 7).

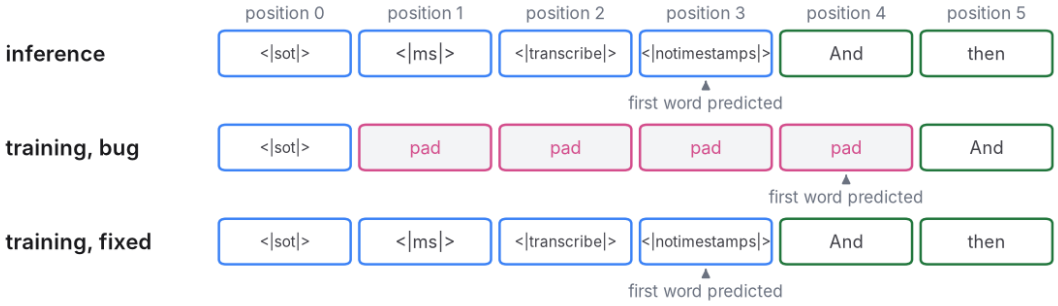


FIGURE 7 Decoder inputs for the same clip. Training with the bug never feeds the language and task tokens; the fix feeds them as inputs and keeps them out of the loss.

The second is in the full fine-tunes: the model is loaded in bfloat16 and trained in it. An AdamW step at a learning rate of 10^{-5} is about 0.05% of a typical weight, and bfloat16 resolves about 0.4%, so most updates round to nothing. Full fine-tunes keep fp32 master weights here (bfloat16 compute through autocast; reentrant gradient checkpointing then has to go, because it recomputes layers against autocast's cached weight casts).

The third is in what a checkpoint saves. Base's generation config forces `<|transcribe|>`; the fine-tunes' came out without it, so a call that names no task left the task token to the model, and a fine-tune that has learned to obey its prompt sometimes chose `<|translate|>` and wrote Yoruba as English. vLLM always states `<|transcribe|>`; so does every number below, and so does the released model's config.

4.2 MEASURED AS IT IS SERVED

Hugging Face's Whisper config forbids `<|endoftext|>` as the first token generated after the prompt. With the prompt stated, that is the first word, so the model cannot return an empty transcript: on a silent clip it must write something, and a fine-tune that has learned to stop writes its second choice. vLLM, which serves the model in the pipeline, forbids nothing. The same checkpoints decoded both ways on the 1,810 voice-free clips (Figure 8):

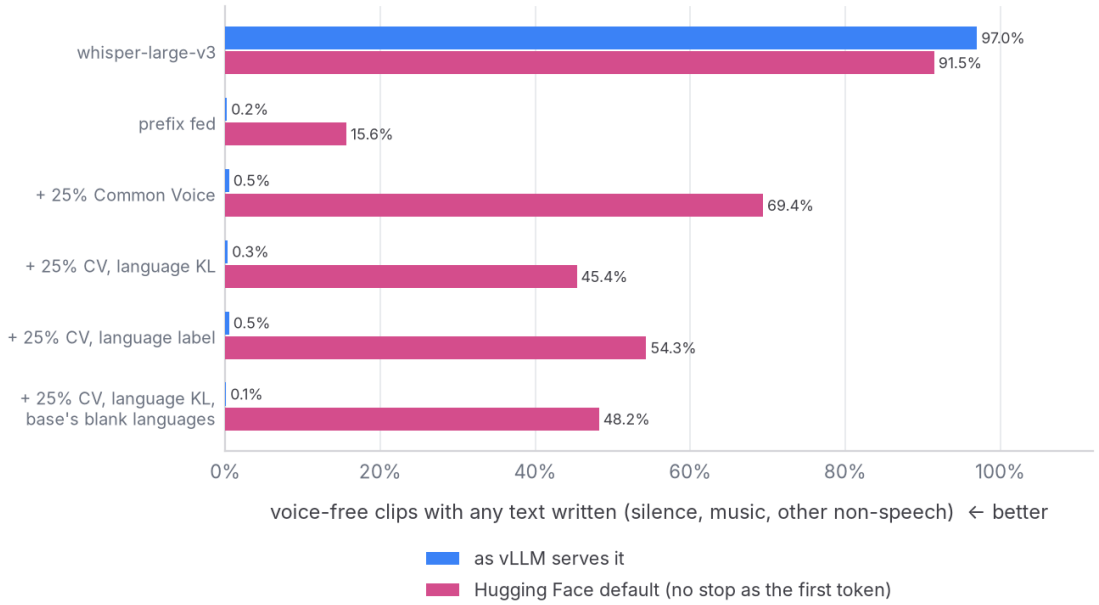


FIGURE 8 Voice-free clips (silence, music, other non-speech) with any text written, decoded with Hugging Face's default (no stop as the first token) and as vLLM serves (nothing forbidden). Under the default the fine-tunes' commonest outputs are a full stop, "you", "Thank you." and "Oh"; as served, nothing.

Base writes over 91.5% of them with the default and 97.0% as served (the difference is music notes, which the default also suppresses): it never wanted to stop. The fine-tune with 25% replay writes over 69.4% with the default and 0.5% as served. With the default every clip gets some output (100%), most often a full stop or "you". Measured that way, replay and the language losses below looked as if they gave back half of the fix; as served, none of them did.

vLLM itself, run through its own detection and transcription prompts, wrote the same transcript as our decoding on 100.0% of the 1,810 voice-free clips, in float16 and in bfloat16. On the 1,160 FLEURS clips it detected the same language on 99.9% and wrote the same transcript on 93.7% in float16 (74.3% in bfloat16, which the pipeline serves); the long transcripts drift with the kernels' rounding, the error rate does not (macro CER 0.274 ours, 0.278 and 0.276 vLLM). Every number from here on is decoded as served.

4.3 HALLUCINATION

As served, the fix holds (Figure 9). The recipe as published still writes over 33.3% of silent clips and 12.8% of real voice-free audio; fed its prefix, the same recipe writes over 0.0% and 0.9%.

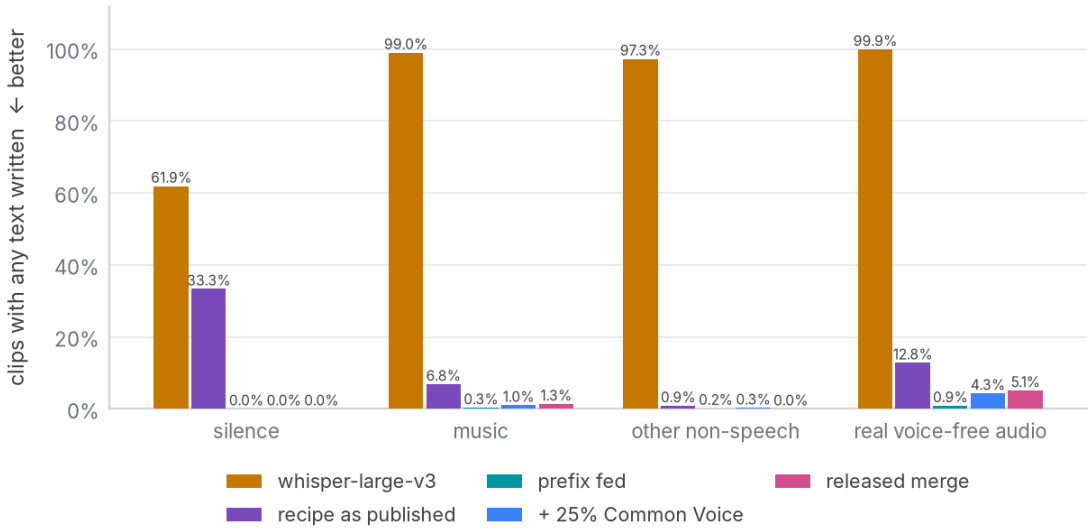


FIGURE 9 Clips with any text written, on four voice-free arms, decoded as served. "Real voice-free audio" is audio from the wild that a VAD confirms has no voice.

4.4 WHAT STOPPING COSTS

A model that has learned to stop can stop on speech. Without replay it stops on whole languages (Figure 10): fed its prefix, the fine-tune writes nothing for 29.6% of the FLEURS clips [Conneau et al., 2023] (1,160 clips in 58 languages, none trained on), and for most of them in Japanese (95%), Shona (95%), Burmese (90%) and Yoruba (90%). Its training speech was Malay, English and the spoken hallucination phrases, so a language it never heard sounds like non-speech. With 25% Common Voice replayed it writes nothing for 0.1%, and no language passes 5%.

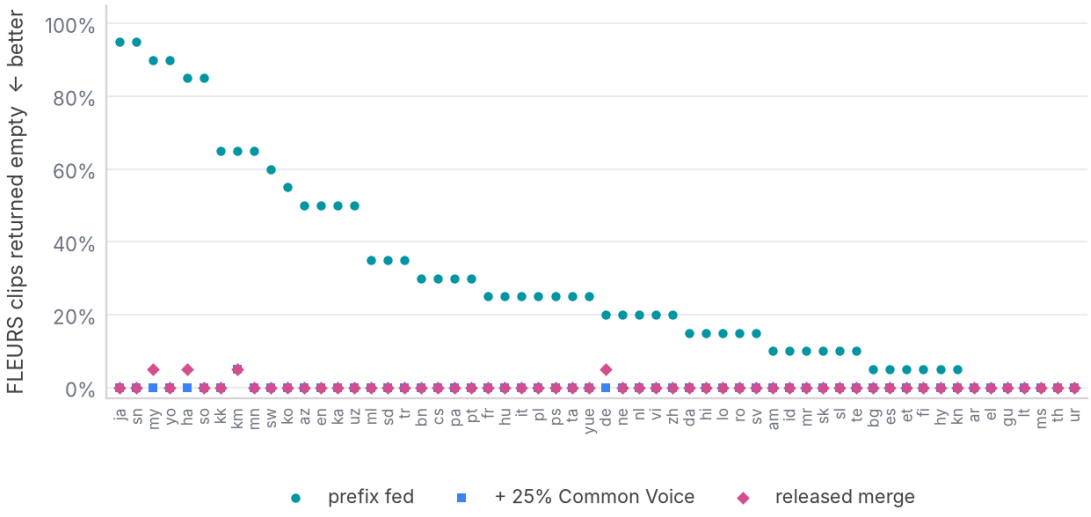


FIGURE 10 FLEURS clips returned empty, per language, without replay and with 25% Common Voice replayed.

This is the multilingual loss the recipe's paper reports, and it is larger than reported: under the default decoding the model could not stop, so it wrote something, often the right thing.

Over every speech arm (Figure 11) replay keeps the empty share near zero. One more choice raised it. We had trained each voice-free clip under the language base detects on it, the token it meets at inference. Base detects English on 76% of them, so the model learned to stop after <|en|>: it returned nothing for 3.8% of LibriSpeech, which is read English.

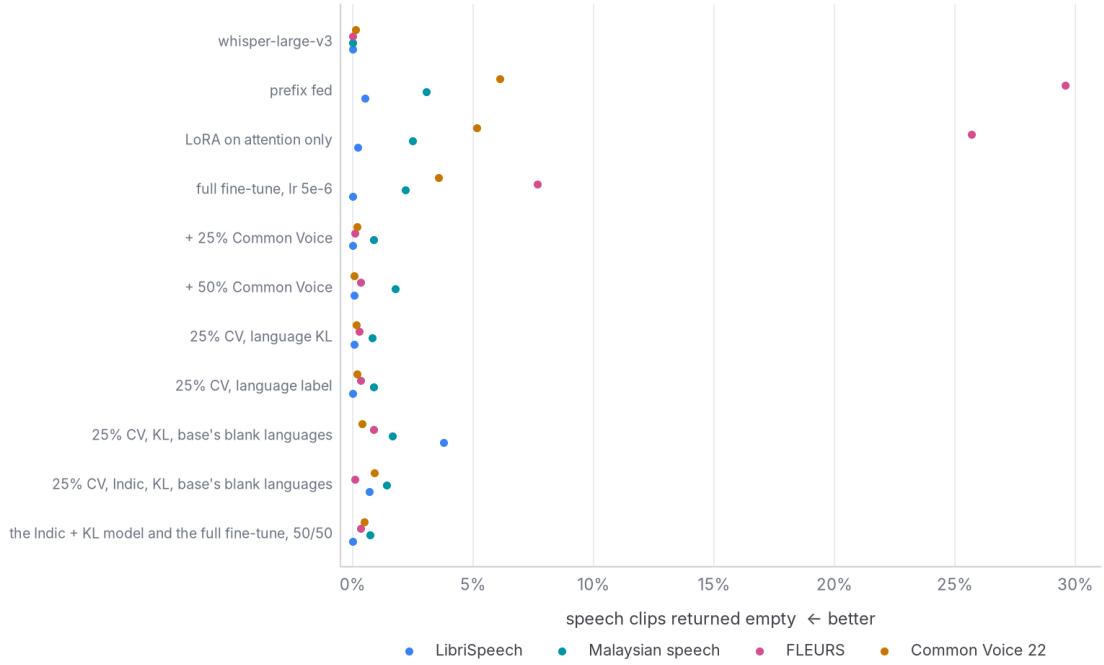


FIGURE 11 Speech clips returned empty, per arm. Base always writes something: it cannot stop as its first token.

Noise is the other place a fine-tune stops (Figure 12). With 25% replay, 0.4% of the speech at 20 dB SNR comes back empty, and 14.2% at −5 dB.

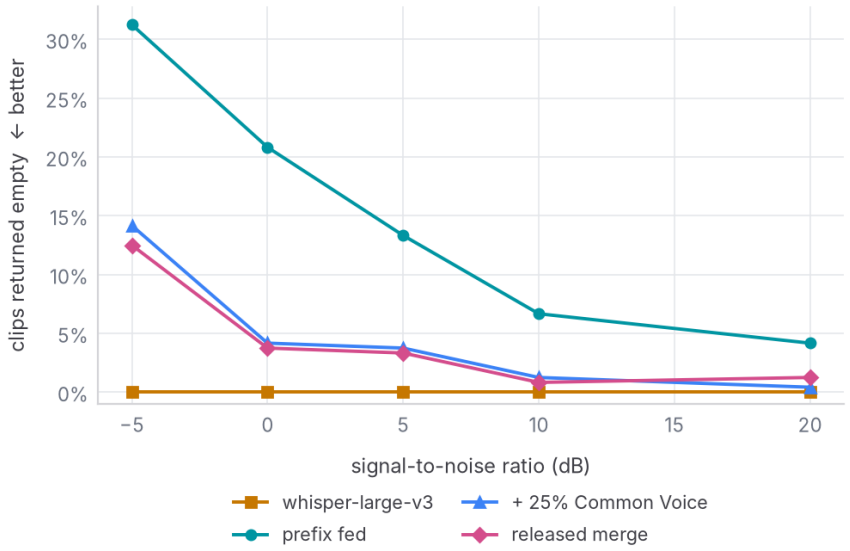


FIGURE 12 Speech under music or noise returned empty, by signal-to-noise ratio.

4.5 THE REST OF THE MULTILINGUAL LOSS IS LANGUAGE IDENTIFICATION

Whisper predicts the language token from the start token alone. Masked out of the loss, that prediction drifts with every LoRA step, and a model that has learned to obey its prompt transcribes into whatever language it guessed. Forcing each clip's language separates the two (Figure 13): forced, the replay models transcribe better than base (0.235 with 25% replay, 0.259 base); detected, they are level with it (0.294, 0.282).

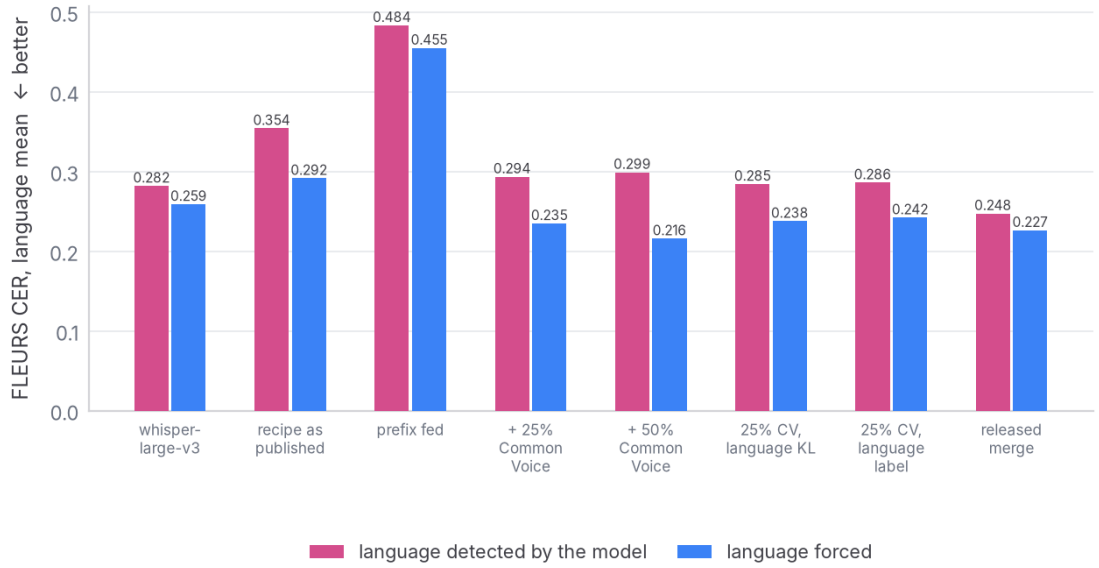


FIGURE 13 FLEURS character error rate, mean over languages, with the language detected by the model and forced to the clip's own.

Replay alone lowers identification, to 72.0% against base's 83.7% (Figure 14). Two losses hold it: a KL term keeping the first token's distribution over language tokens at base's (82.3% with replay), and supervising the language token on speech (87.7%).

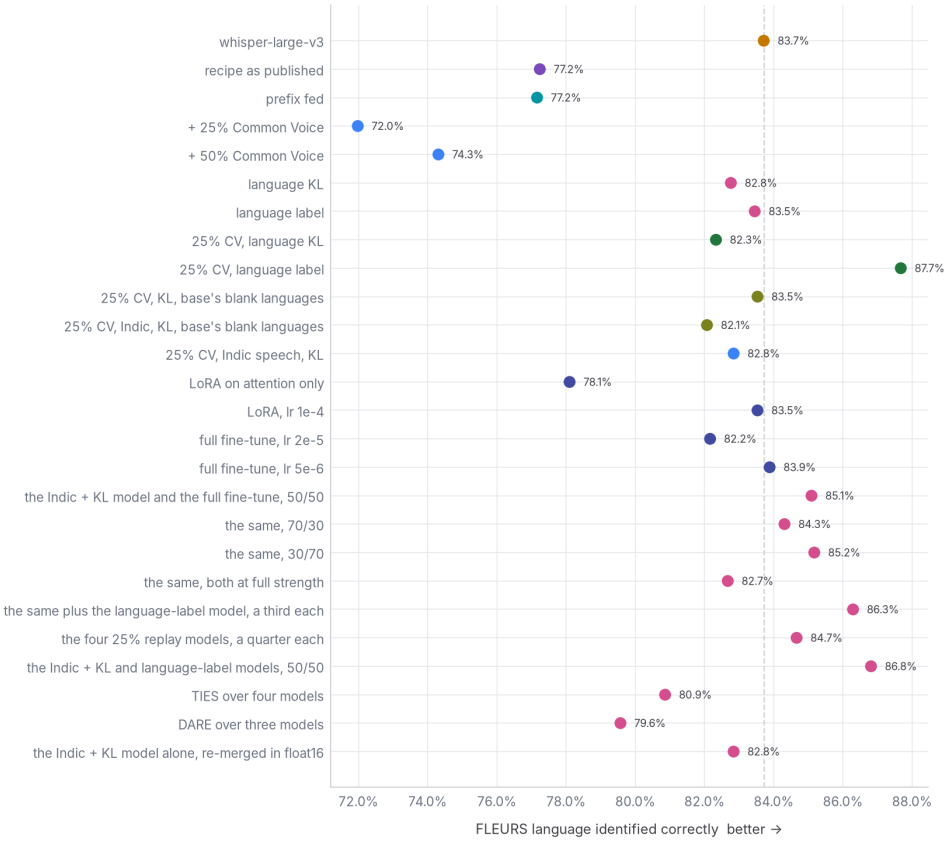


FIGURE 14 FLEURS clips whose language the model identifies, every checkpoint. The dashed line is base.

Clips that fail outright (CER of 0.8 or more) tell the same story (Figure 15).

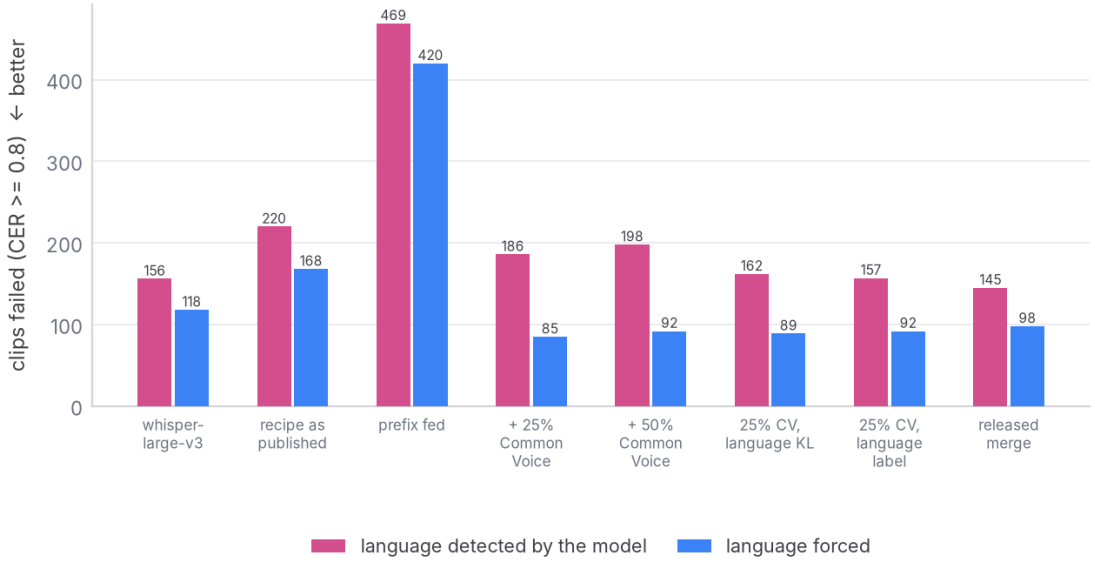


FIGURE 15 FLEURS clips with a CER of 0.8 or more, out of 1,160.

4.6 LORA OR ALL PARAMETERS

The recipe's paper chose LoRA on the attention and MLP projections at 5×10^{-4}. On its mix, without replay, every variant treats some languages as non-speech (Figure 16): the full fine-tunes least (7.7% of FLEURS at 5×10^{-6}), LoRA on attention only nearly as much as the recipe. Replay matters more than the method.

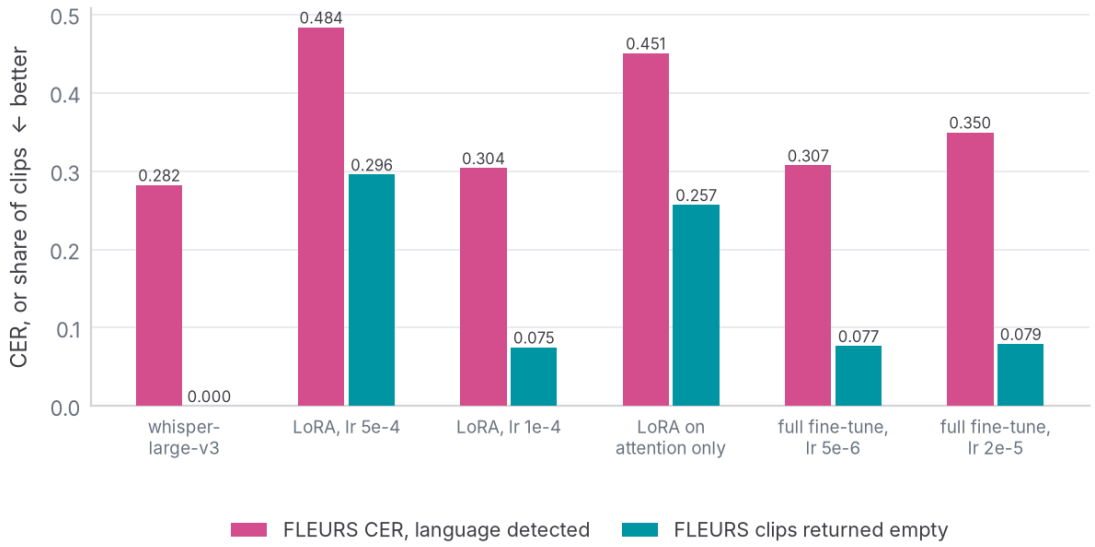


FIGURE 16 The same mix, no replay, five ways of training: FLEURS CER with the language detected, and the share of FLEURS clips returned empty.

4.7 LOOPS

Asked to transcribe a syllable said several times, the replay fine-tunes sometimes cannot stop (Figure 17): for "ta" said eight times, the model with 25% replay and the language KL wrote "ta" until its 440-token limit. The loops were there before the decoding changed. Melati's ASR stage guards against them the way Whisper's own decoder does: a transcript that compresses like a loop (gzip ratio over 2.4) is decoded again at temperatures 0.2, 0.4 and 0.6, and every decode is capped at 15 tokens per second of audio.

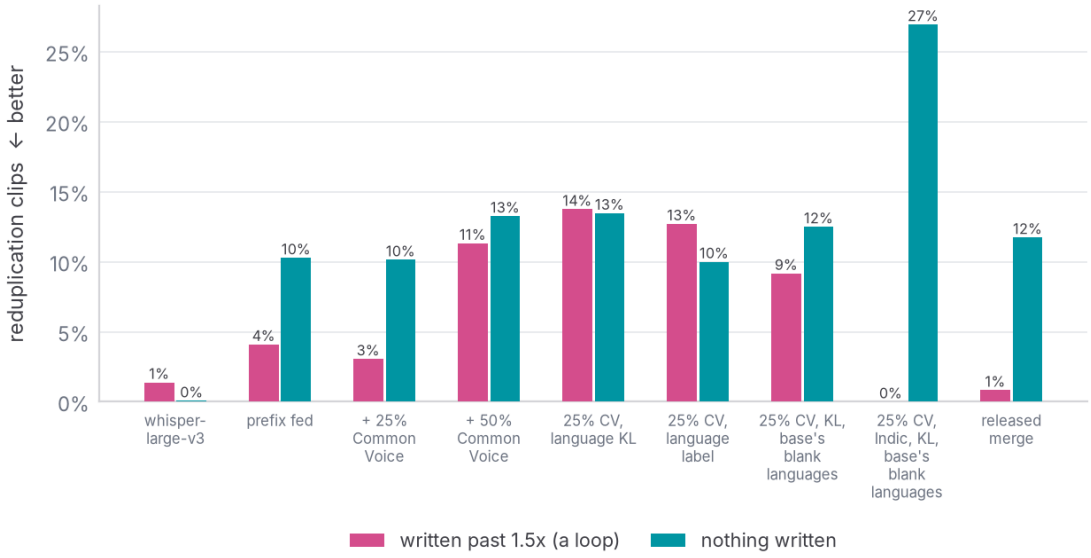


FIGURE 17 The reduplication arm (a syllable said *n* times): clips where it is written more than 1.5*n* times, and clips where nothing is written.

4.8 MERGING

Each fine-tune did one thing well. Replay with the Indic arm and the language KL kept the languages; the full fine-tune at 5×10^{-6} barely loops and has the best LibriSpeech WER; supervising the language token held identification best. All are changes to the same base, so they can be added [Wortsman et al., 2022, Ilharco et al., 2023]. We rebuilt every LoRA change exactly from its fp32 adapter (the merged checkpoints were saved in bfloat16; re-merging the Indic + KL model alone in float16 gave the same numbers), averaged the changes with weights, and scored each merge on the arms that separate the fine-tunes (Table 2).

	loops	LibriSp.	Malaysian	FLEURS	FL. empty	lang. ID
whisper-large-v3	1.4%	3.06%	23.4%	0.282	0.0%	83.7%
Indic + KL	4.3%	3.38%	22.8%	0.256	0.0%	82.8%
full fine-tune, 5×10^{-6}	0.8%	2.89%	22.3%	0.307	7.7%	83.9%
the two, 50/50 (released)	0.8%	3.09%	21.6%	0.248	0.3%	85.1%
the two, 70/30	1.4%	3.22%	21.5%	0.247	0.1%	84.3%
the two, 30/70	0.5%	2.95%	21.6%	0.261	0.9%	85.2%
the two, both at full strength	4.8%	3.77%	24.0%	0.294	3.2%	82.7%
the two + label model, a third each	3.6%	2.91%	21.8%	0.248	0.1%	86.3%
four replay models, a quarter each	12.4%	2.90%	21.5%	0.246	0.1%	84.7%
TIES, four models	6.8%	3.55%	23.2%	0.277	0.0%	80.9%
DARE, three models	13.8%	6.11%	35.7%	0.509	0.8%	79.6%

TABLE 2 Merges of the fine-tunes' weight changes, decoded as served. Loops: reduplication clips written past 1.5×. WER is over every reference word; FLEURS is CER with the language detected.

Averaging the Indic + KL model with the full fine-tune, half each, beat both on Malaysian speech and FLEURS, kept language identification above base and cut the loops to 0.8%.

Shifting the weights trades loops against FLEURS clips returned empty; adding both changes at full strength, TIES [Yadav et al., 2023] and DARE [Yu et al., 2024] were worse, and averaging the four replay models brought the loops back.

4.9 THE RELEASED MODEL

The released model is that half-and-half merge, at `https://huggingface.co/Scicom-intl/Melati-Whisper-Large-V3` (Table 3). Its generation config suppresses nothing and states the transcribe task, so `transformers` decodes it as vLLM does.

	whisper-large-v3	released
Silence, music, other non-speech with text	61.9%, 99.0%, 97.3%	0.0%, 1.3%, 0.0%
Real voice-free audio with text	99.9%	5.1%
Spoken hallucination phrases transcribed	67.9%	73.2%
FLEURS CER, language detected / forced	0.282 / 0.259	0.248 / 0.227
FLEURS language identified	83.7%	85.1%
Common Voice 22 CER	0.353	0.307
LibriSpeech test-clean WER	3.06%	3.09%
Malaysian speech WER	23.4%	21.6%
Returned empty: LibriSpeech, Malaysian, FLEURS	0.0%, 0.0%, 0.0%	0.0%, 0.7%, 0.3%
Reduplication: loops, nothing written	1.4%, 0.1%	0.8%, 11.7%

TABLE 3 The released model against base, decoded as served. WER is over every reference word.

Language by language (Figure 18) it sits on or under base for most of FLEURS.

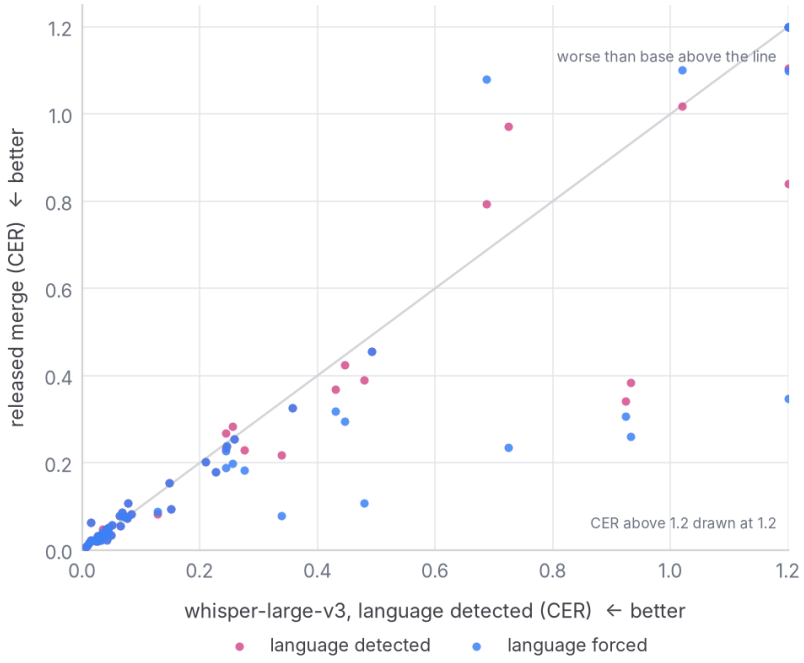


FIGURE 18 FLEURS CER per language, base (x) against the released merge (y), with the language detected and forced.

5 Tags with measured precision

A tag in a training transcript is an instruction to a speech generator, so a wrong tag is worse than a missing one. Every tag class here is kept only where its precision was measured on held-out labels, and the threshold is the one that measurement chose.

5.1 EMOTION

We started from two released classifiers: MERaLiON-SER [Sailor et al., 2025], trained for English, Chinese, Malay and Tamil, and emotion2vec+ large [Ma et al., 2023]. emotion2vec+ was fine-tuned on EmoBox [Ma et al., 2024], whose 32 corpora include 12 of CAMEO's thirteen [Christop and Czajka, 2025], and on those it is near perfect (Figure 19). Only corpora outside every model's training data say anything: nEMO [Christop, 2024] (Polish, acted) and the VoxCeleb half of NonverbalTTS [Borisov et al., 2025] (English interviews), 7,791 clips.

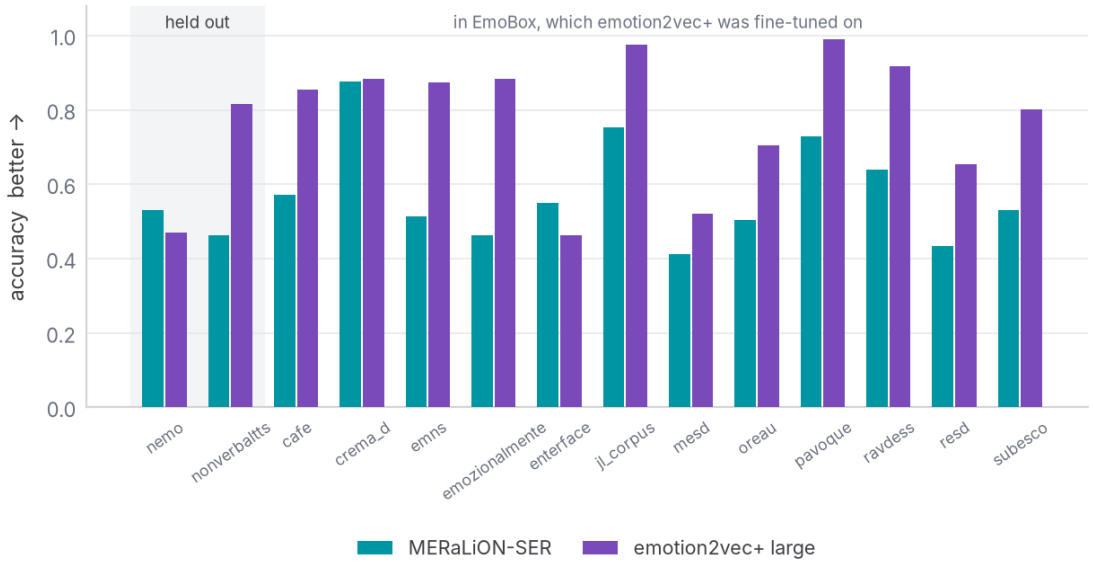


FIGURE 19 Accuracy per corpus. The shaded corpora are held out from every model we know the training data of.

Our own heads. We trained a small head (a learned mix of the encoder's layers, mean-pooled over the clip, then one hidden layer) on five frozen encoders: Whisper-large-v3 [Radford et al., 2023], the strongest backbone in EmoBox's benchmark; Melati's own Whisper fine-tune; w2v-BERT 2.0 [Seamless Communication et al., 2023]; WavLM-large [Chen et al., 2022]; and emotion2vec+'s features. The training set is every corpus of EmoBox we could download without a signed agreement, CAMEO, Expresso and the Expresso half of NonverbalTTS: 67,841 clips from 21 corpora in 11 languages, with every corpus weighted equally in the loss. It includes 13,706 clips of MELD [Poria et al., 2019], TV dialogue, the nearest thing to conversation that is openly licensed. On the held-out clips the head on Whisper-large-v3 beats both released classifiers alone and averaged (Figure 20); averaging all seven models does better still, 0.597 against 0.529.

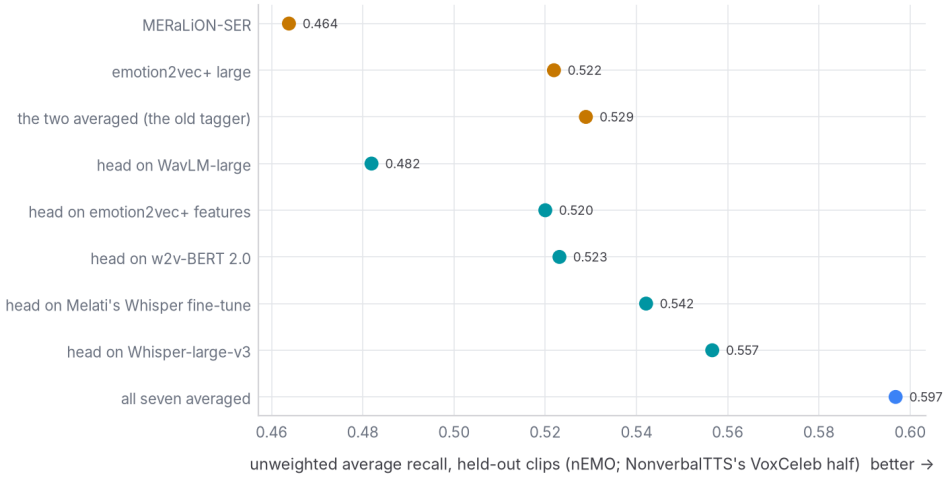


FIGURE 20 Unweighted average recall on the 7,791 held-out clips, every model scored on the same clips. The seven are averaged as they are, not picked after the fact.

When a label is tagged. Acted corpora are balanced; podcasts are mostly neutral. A threshold with 0.8 precision when a seventh of the clips are angry has far less when one in twenty is. So precision is re-weighted to an assumed podcast rate per label (happy 15%, sad, angry and surprised 5%, fearful and disgusted 2%), and a label is tagged only where that precision reaches 75%. The threshold is chosen on one half of the held-out clips and its recall measured on the other (Figure 21). The old tagger could tag happy, angry and surprised; the seven together also tag sad (49%) and fearful (29%), and find more of the happy clips (59% against 47%). Disgust has no held-out examples and is never tagged. Every label's probability stays in the dataset.

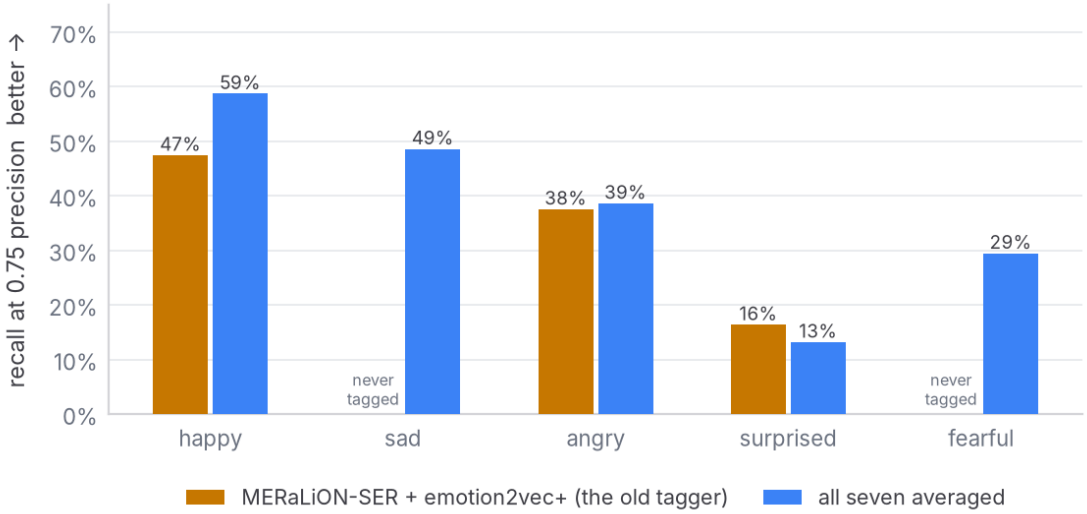


FIGURE 21 Recall at 0.75 precision, re-weighted to podcast rates, held-out clips; the threshold is chosen on one half and measured on the other.

Mandarin film and TV. MER2023 [Lian et al., 2023], licensed for academic research, adds 5,030 labelled clips of Mandarin film and television, "worried" folded into fearful as anxiety is. Left out of training, the head on Whisper-large-v3 scores 0.48 UAR on it (emotion2vec+'s

features, which saw it during their own training, 0.59) and hears 13% of the worried clips as fearful. Trained in, it moves the held-out average by less than one seed would (Figure 22): 0.590 against 0.597, nEMO 0.602 against 0.608, NonverbalTTS 0.803 against 0.802. The tagger stays as it was.

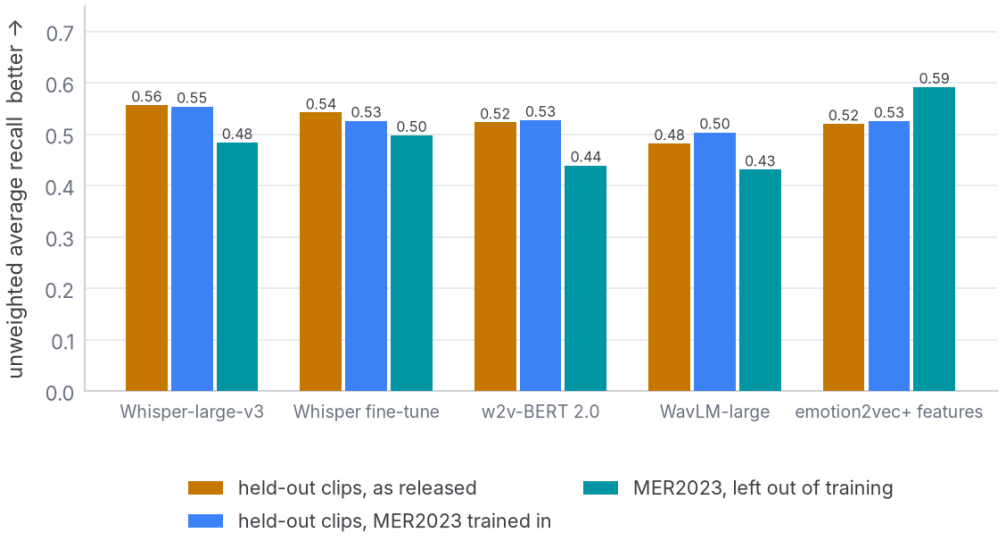


FIGURE 22 Each head on the held-out clips, as released and with MER2023 trained in, and on MER2023 when it is left out of training.

None of these held-out clips is Malaysian. 668 segments of the podcast pilot, 400 of them drawn at random to measure how often each emotion really occurs, are ready for native speakers to label on a page that keeps the audio on our server; until they are, the precision on Malaysian podcasts is assumed.

5.2 NON-VERBAL SOUNDS

NonverbalTTS marks vocalisations inline, at the word boundary where they happen; its test split has 492 breaths, 56 laughs and 29 coughs in 359 clips. A tag counts when it lands within one word of a reference tag of its class. We tried three ways.

An audio event detector, PANNs, placed at the nearest word boundary: precise and nearly blind (laughs 100% precision, 10.7% recall; breaths 0.3% recall). An audio language model asked to transcribe and tag in one pass: Qwen3-Omni finds more, and its wording decides how much (with a prompt that asks for every breath, breath recall rises to 55%), but its tags land at the wrong word (13% of its breath tags within one word) and it invents throat clearing (222 clips with none).

So we split the question (Figure 23). Where: the aligned words leave gaps, and a gap with sound above both the clip's noise floor and a level 40 dB under its speech is a candidate, with its own times. What: each candidate, cut out and padded with silence, goes to Qwen3-Omni with a closed list of sixteen labels as letters (speech, filler, breath, laugh, cough, sigh and so on, silence and noise included), and the probability of each letter is read from the first answer token. A class is tagged where its probability clears a threshold chosen on 800 NonverbalTTS training clips for 80% precision; the test split takes no part in that choice.

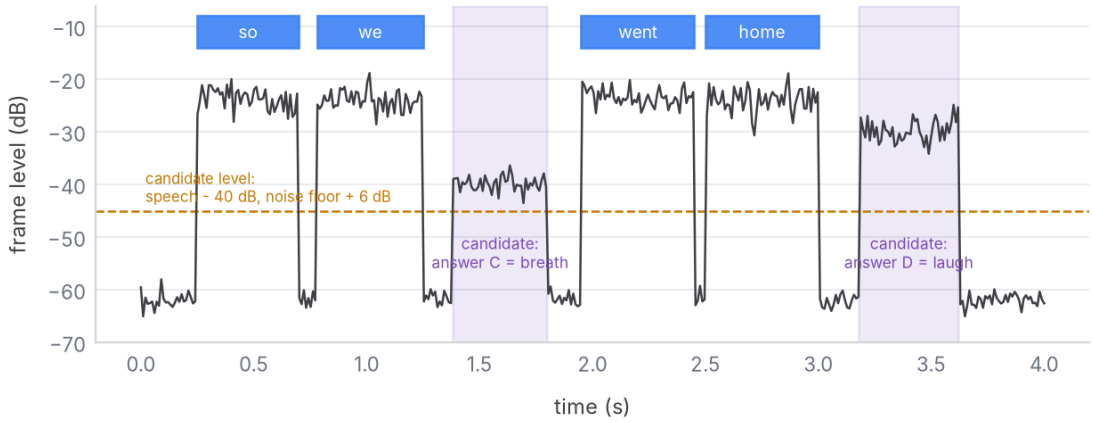


FIGURE 23 Where by acoustics, what by an audio language model. Illustration on a synthetic signal.

Figures 24, 25 and 26 compare the approaches on the test split; the last of them, our own classifier, is described below.

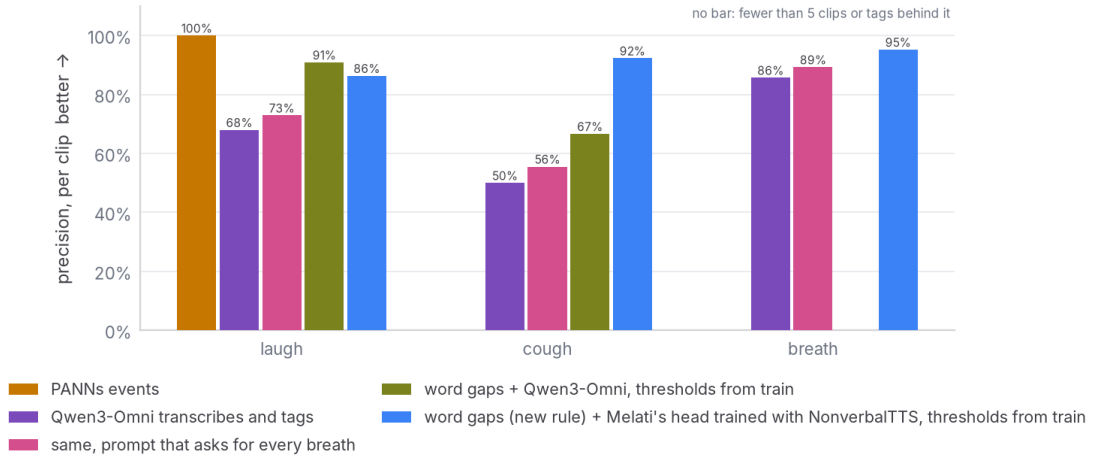


FIGURE 24 Precision: of the clips a method says contain the sound, the share that do.

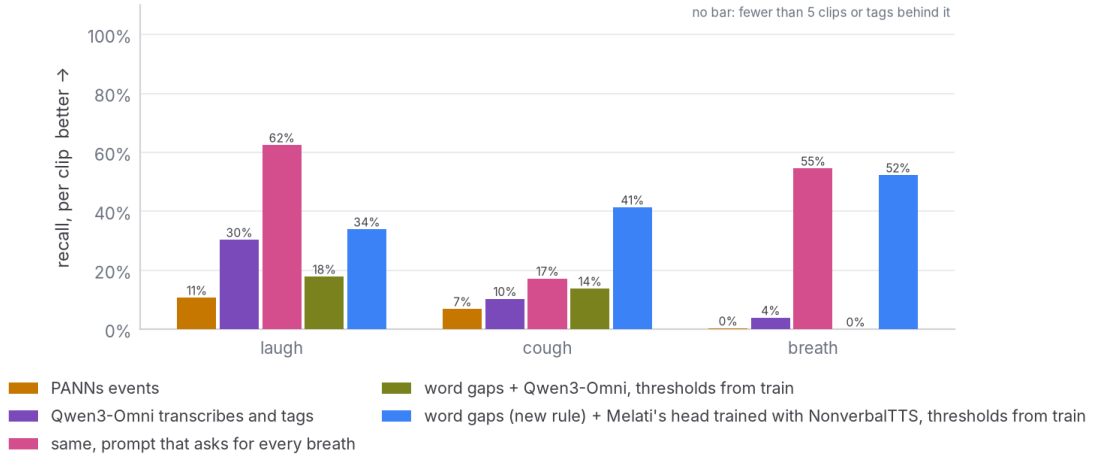


FIGURE 25 Recall: of the clips that contain the sound, the share a method finds.

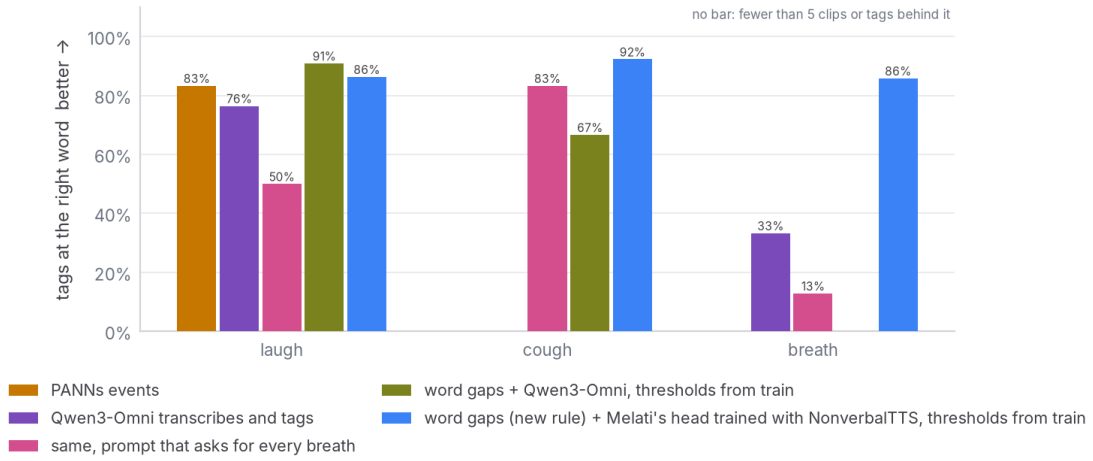


FIGURE 26 Tags at the right word: the share of a method's tags within one word of a reference tag of their class.

Qwen3-Omni tags laughs at 91% precision (threshold 0.05) and coughs at 67% (0.95). Recall is the cost: 18% of laughs. A laugh over speech is not in a gap, but most are: 77% of the reference laughs have a candidate at their boundary, so it is the labeller that passes most of them up. Its breaths never reach 80% on the training clips (Figure 27), so it tags none.

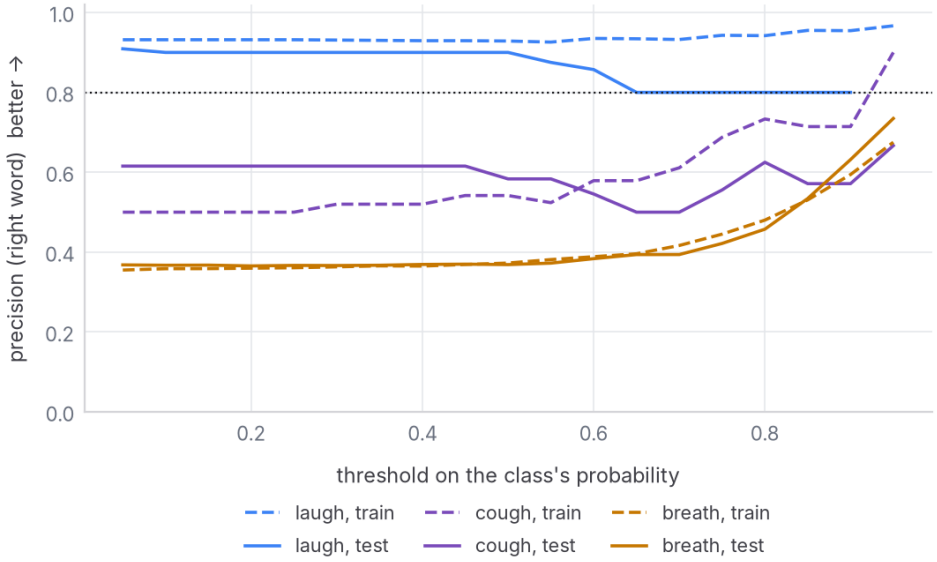


FIGURE 27 Tag precision against threshold, training clips (dashed) and test clips (solid). The dotted line is the bar for tagging.

Our own classifier. So we replaced the labeller with the emotion heads' design (a learned mix of a frozen encoder's layers, mean-pooled over the clip, one hidden layer) and an output for each of speech, filler, breath, laugh, cough, sigh, sniff, throat clearing, sneeze, noise and *other*, a gap sound no annotator tagged. Trained on hand-cut events alone, the training episodes of PodcastFillers [Zhu et al., 2022] (English podcasts with every filler, laugh and breath timed by hand) and VocalSound's [Gong et al., 2022] recordings, the head on Whisper-large-v3 tells held-out events apart (macro F1 0.89) and real candidates badly: averaged with heads on WavLM-large and AST [Gong et al., 2021], its laughs on NonverbalTTS are 64%

precise and its breaths almost never clear their threshold. A candidate is whatever an energy rule cuts out of a word gap, mostly word tails and clicks that no one cut by hand. So the training set gained the candidates themselves: 48,205 from PodcastFillers' training episodes, found by the pipeline's own rule over the word times its transcripts carry and labelled by the hand-timed event each overlaps, and 6,364 from NonverbalTTS training clips outside the 800 that choose the thresholds, labelled by the tag at their boundary or as *other*; 141,242 clips in all (Figure 28).

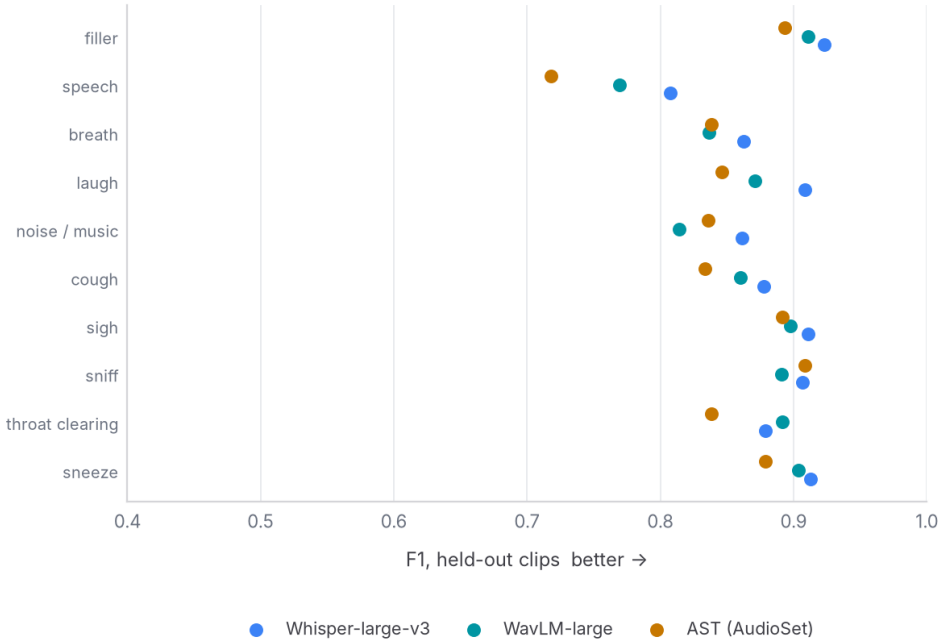


FIGURE 28 Each head on events nothing trained on: PodcastFillers' test episodes (shows no model here saw) and VocalSound's test speakers. F1 at the most probable class.

On the same NonverbalTTS test candidates, with thresholds chosen the same way, the head on Whisper-large-v3 tags 46% of the laughs at 81% precision (Qwen3-Omni 18% at 91%), 17% of the breaths at 87%, and 28% of the coughs at 100% (Figures 24 to 26). The three heads averaged do worse (laughs 75% precise), so the pipeline runs the Whisper head alone: one encoder on the tag server where Qwen3-Omni took a GPU of its own.

A finder for quiet breaths. The rest of the breaths were lost before any labeller saw them: only 54% of the test breaths had a candidate. Hardly any of the others lack a gap (5 of 492); they sit in real gaps, too brief above the level threshold (111) or under it (110), by a median 1.7 dB (Figure 29). The threshold that bound was the noise floor's, never the speech level's: these are interview recordings, and a breath is barely above the room. With the floor margin at 2 dB instead of 6, 91% of the test breaths have a candidate, at 3.6 candidates per clip instead of 2.4. The head was retrained on what the looser rule finds, and the rule chosen on the calibration clips alone; it tags 35% of the test breaths at 88% precision, 59% of the coughs at 85%, and 45% of the laughs at 86% (Figure 30).

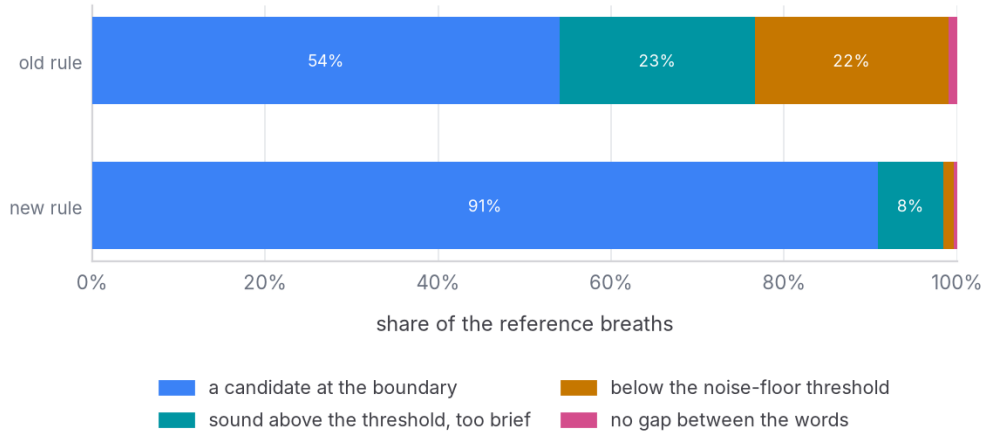


FIGURE 29 The NonverbalTTS test breaths, by what the gap finder made of them: a candidate within one word, sound in the gap above the threshold for under 80 ms, sound under it, or no gap between the words at all.

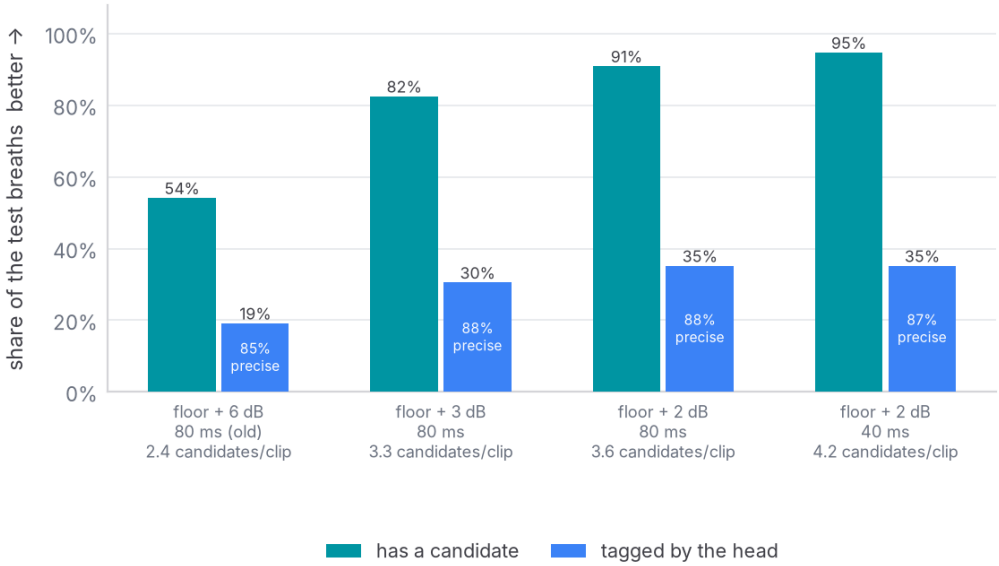


FIGURE 30 Each candidate rule on the NonverbalTTS test breaths: the share with a candidate, and the share tagged by the head retrained on the loosest rule's candidates, at thresholds chosen on the calibration clips.

The released model. PodcastFillers' annotations are licensed for non-commercial research only, and the sources beside them must allow that in one set of weights: VocalSound is CC BY-SA, and so are 79 of PodcastFillers' own 199 episodes, with 20 more CC BY-ND. NonverbalTTS is the third: its metadata says Apache-2.0, but its README licenses the annotations CC BY-NC-SA, so the head above, which trained on its candidates, cannot be released. Without them, a head trained on PodcastFillers' CC BY episodes and COUGHVID's coughs [Orlandic et al., 2021] must raise its thresholds to 0.95 to reach 80% precision on NonverbalTTS's training clips, and then finds 4% of the test laughs: those interviews are a domain of their own, which only their own candidates taught. So the released model is calibrated where Melati works, on podcasts: its breath threshold is the lowest giving 80% precision where PodcastFillers' validation episodes have a label (0.70). Its gap head

trains on PodcastFillers' CC BY episodes, AMI's meetings [Carletta et al., 2005] (CC BY 4.0: 2,563 gap candidates its transcribers marked and 19,205 they did not, as *other*), and the coughs of COUGHVID and Coswara [Bhattacharya et al., 2023] (CC BY 4.0). It tags breaths only. Its own laugh tags in podcasts come at twice the first head's rate for the same recall, and AudioSet's classifier hears the extra ones as gasps: 49% of its unlabelled laugh tags have Gasp as their top class, and they score 0.003 on Laughter against 0.015 for hand-labelled laughs. On NonverbalTTS, now out of domain, it tags 10% of the breaths at 69% precision (the first head, trained there: 37% at 86%). Laughs and coughs come from a second head, below. Both are released as one checkpoint at `https://huggingface.co/Scicom-intl/Melati-NonVerbal-Whisper-Large-V3` for non-commercial research.

Laughs and coughs over speech. A gap finder only looks between words. In AMI's test meetings, 74% of the 763 marked laughs and 56% of the 78 coughs lie over someone's words with no candidate at all, and the first head found 4% of the laughs and none of the coughs. A frame-level head reads the whole segment instead: eight layers of the same frozen encoder averaged to ten frames a second, a learned mix of them, and four residual dilated convolutions over time (about 6 s of context), giving every 0.1 s an independent probability of laughter, a cough, another vocal sound and speech. AMI marks every word and every laugh of its conversations, so there a frame with no mark is a negative; PodcastFillers' CC BY training episodes add podcasts, their classes known only inside the regions its annotators labelled. A run of frames over the threshold is an event: laughs at 0.90, chosen on PodcastFillers' validation episodes, and coughs at 0.999, chosen on AMI's dev meetings, since PodcastFillers labels none. It finds 90% of the 579 hand-timed laughs in PodcastFillers' test episodes, 84% precise where labelled; 64% of AMI's test laughs at 65% (65% of those over speech) and 35% of its coughs at 84%; and, out of domain, 50% of NonverbalTTS's laughs at 74% (Figure 31). NonverbalTTS's coughs it does not find: it learnt coughs from meetings, and adding COUGHVID's and Coswara's phone recordings to its training lowered AMI's to 17% without reaching NonverbalTTS's. The pipeline takes laughs and coughs from this head, adding an event wherever the gap head tagged none of its kind within 0.3 s.

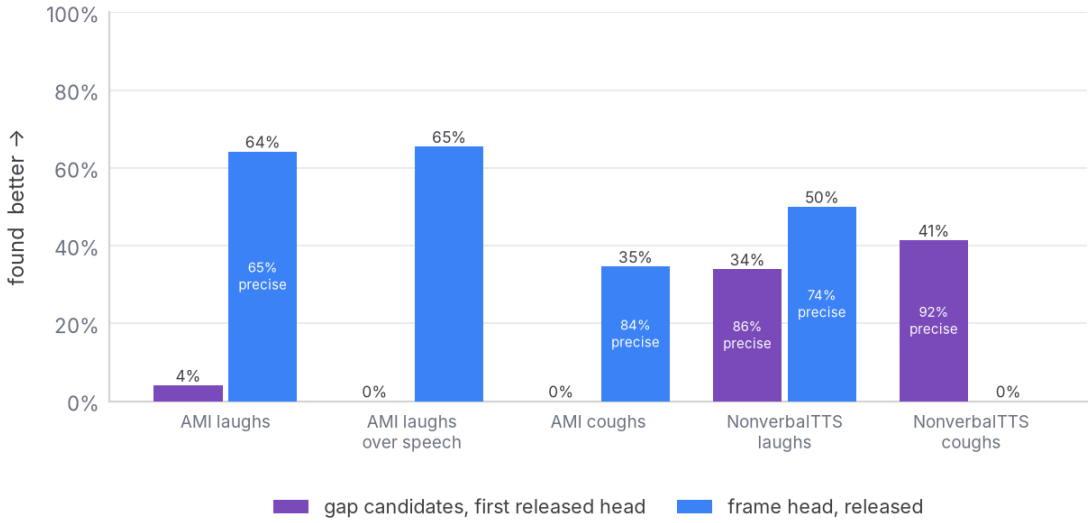


FIGURE 31 Laughs and coughs found by the first released head on gap candidates and by the frame-level head, each at the thresholds it ships with; precision written in the bar. AMI's test meetings: a tag is right where the transcribers marked its class within 0.1 s. NonverbalTTS's test split: within one word of a reference tag.

On PodcastFillers' test episodes, through the whole pipeline (Figure 32), the released model finds 70.7% of the laughs inside what Melati keeps (the first head 59.9%, Qwen3-Omni 46.8%) and 70.1% of the breaths (the first head 49.7%). As scored, 52.0% of its laugh tags and 18.8% of its breath tags are right, but PodcastFillers labels only the regions its own detector proposed, and 1,698 of the 2,202 breath tags fall where it has no label at all. Where it has one, 77% of the laugh tags (Qwen3-Omni's 91%) and 83% of the breath tags are right. The unlabelled breath tags sound to AudioSet's own classifier like hand-labelled breaths: 85% have a gasp, a sigh or a snort as its top class, against 78% of hand-labelled breaths and 4% of hand-labelled words. The looser gap finder is what adds those: the old rule, with a head trained on VocalSound, made 1,389 breath tags, 1,037 of them unlabelled, and found about as many of the breaths PodcastFillers did label as the first released head under the new rule (51.7% against 49.7%): the breaths its detector proposed are the louder ones, which the old rule already reached. The released model's gap head, calibrated on podcasts, finds more of them at the same precision. On the 200 Malaysian podcasts of Section 6 the export carries 63 breath tags an hour (the first head 42, 35 with the old rule; Qwen3-Omni tagged none) and 38 laugh tags (the first head 32, Qwen3-Omni 23), and no cough tags: the frame head's cough threshold, chosen on AMI's meetings, is never reached there (the first head tagged 0.2 an hour). Coughs remain the class no licence-clean source measures in podcasts.

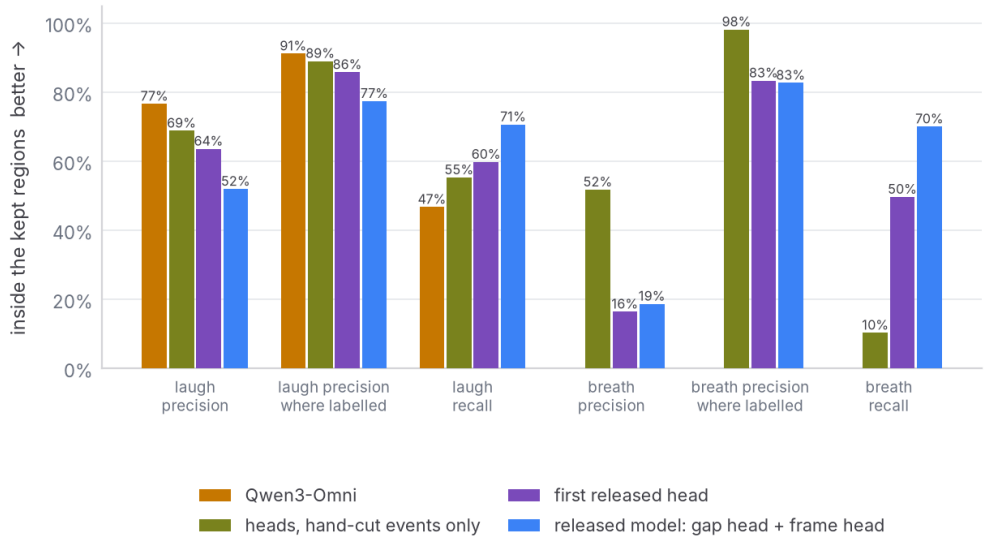


FIGURE 32 PodcastFillers test episodes through the pipeline, inside the regions Melati exports: the same gap candidates labelled by Qwen3-Omni, by heads trained on hand-cut events only and by the first released head, each at thresholds chosen on NonverbalTTS training clips; and the released model, breaths from its gap head and laughs from its frame head, at thresholds chosen on PodcastFillers' validation episodes. Precision as scored counts a tag where PodcastFillers has no label as wrong; where labelled counts only the tags that overlap one of its labels.

Melati v1 tags laughs at 0.97, the threshold AMI's dev meetings choose: on PodcastFillers' test episodes through the pipeline it finds 60% of the laughs, 87% precise where labelled, where the 0.90 the head ships with finds 71% at 77%. The first released head (60% at 86%) and Qwen3-Omni (47% at 91%) lie on the same curve (Figure 33): their higher precision is a stricter threshold, which this head reaches too. A wrong laugh teaches a voice to laugh at the wrong moment, where a missing one only teaches less, so the dataset takes the stricter threshold, and adds the laughs AudioSet's taggers hear (below).

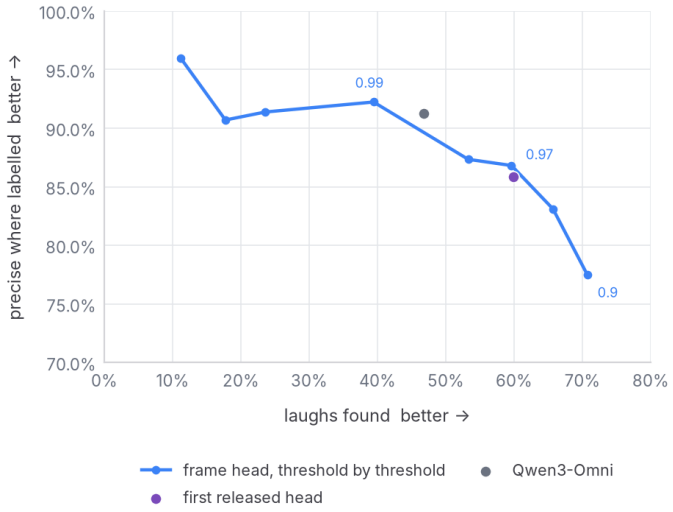


FIGURE 33 Laughs in PodcastFillers' test episodes through the pipeline, at every threshold of the frame head; the first released head and Qwen3-Omni at the thresholds they ship with. Precision counts only tags where PodcastFillers has a label.

AudioSet's taggers. Three of the frame-level models Schmid et al. [2025] fine-tuned on AudioSet's time-stamped labels [Gemmeke et al., 2017], BEATs, ATST-F and M2D (447 classes every 40 ms), averaged frame by frame, hear sounds no head here was trained for. On NonverbalTTS's 3,970 clips (10.3 hours, none of which they trained on), with each threshold chosen on one half for 80% precision and measured on the other, sniffs at 0.15 are 85 to 88% precise and 68 to 71% found, throat clearing at 0.20 is 81 to 83% precise and 34 to 47% found, and coughs at 0.50 are 80 to 97% precise and 17 to 28% found, where the frame head finds none of NonverbalTTS's coughs (Figure 34). On Expresso they mark 99% of the whispered utterances and 0% of the 10,097 in its other styles, so whispering is a delivery tag over the words it covers. Their laughs at 0.30, added to the frame head's at 0.97, find 71% of PodcastFillers' test laughs through the pipeline, 84% precise where labelled: as many as the head alone at 0.90 (71%), at its stricter precision. Sneezes and groans hold 80% on neither half, gasps land on labelled breaths and laughs, breathing fires on every sound, and sighs, crying, humming, singing and shouting are almost never heard in the podcasts, so none of them is tagged.

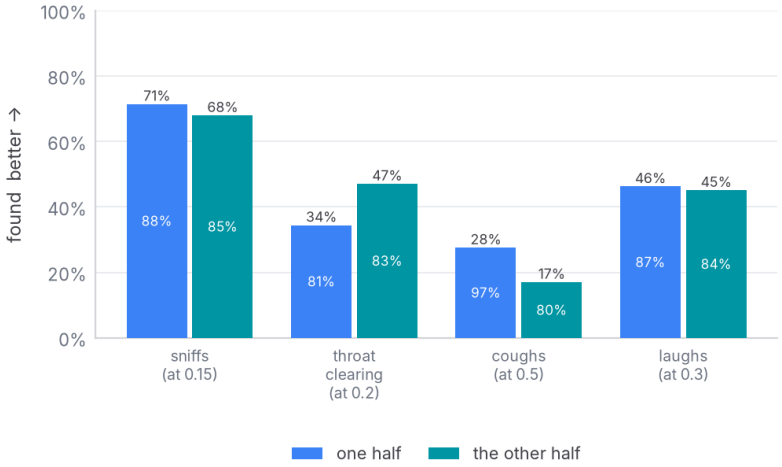


FIGURE 34 AudioSet's taggers on NonverbalTTS at the thresholds Melati v1 tags at: found in each half of the clips (over the bar), each threshold chosen on the other half for 80% precision; precision written in the bar. A tag is right within one word of a reference tag of its class.

5.3 FILLERS

A filler reaches the transcript two ways: Whisper writes it ("um", "erm"), or stage H transcribes a voiced gap on its own and the result is a filler. Stage H depends on the fine-tune: a model that writes text over a breath would turn every gap into a word.

We scored them on PodcastFillers [Zhu et al., 2022], English podcasts with every "uh" and "um", laugh and breath timed by hand. Melati ran on its 20 test episodes (shows no model here trained on) twice, once with base whisper-large-v3 and once with the released model (Section 4), everything else the same; a tag counts when it lands within 200 ms of a label of its class (Figure 35).

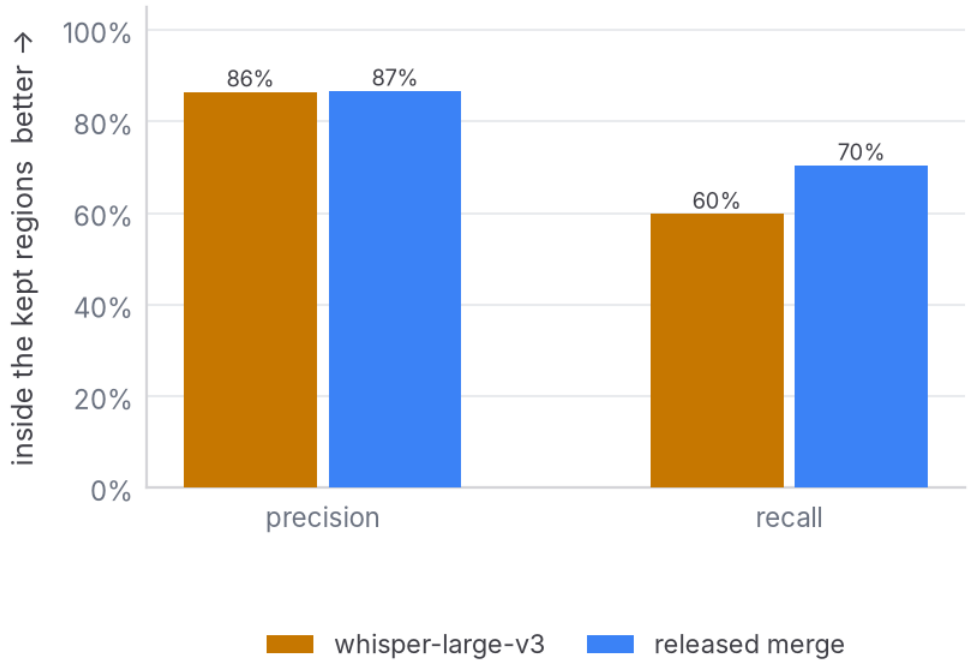


FIGURE 35 PodcastFillers test episodes, inside the regions Melati exports: the fillers it wrote, against the hand labels, with base Whisper and with the released model.

With base, Melati finds 59.8% of the fillers inside what it keeps, at 86.4% precision; with the released model, 70.4% at 86.6%: it writes the fillers base leaves out, at the same precision. Laughs and breaths are the non-verbal head's (Figure 32). Over whole episodes filler recall is lower (59.7% of 5,044 fillers), because export leaves out what it cannot keep cleanly: overlapped speech, music, the edges of a cut.

5.4 AN AUDIOSET TAGGER

The dataset's AudioSet tags come from PretrainedSED's frame-level models (above). For tagging whole clips we also trained a tagger of our own, and released it. First we scored six released clip taggers on the same clips: the 17,141 clips of AudioSet's evaluation set that agkphysics/AudioSet holds today. YouTube removes clips over time, so every paper's copy differs and published numbers do not compare across papers. Each model ran in fp32 with its own preprocessing. CED-base [Dinkel et al., 2024a] leads at 0.4997 mAP, ahead of EAT-large [Chen et al., 2024] at 0.4947 and SSLAM [Alex et al., 2025] at 0.4944, and the average of those three reaches 0.5166 (Figure 36). BEATs [Chen et al., 2023] scores 0.4835. bf16 lowers each tagger's score by at most 0.1 of a point, so training evaluates in bf16 and every number reported is fp32.

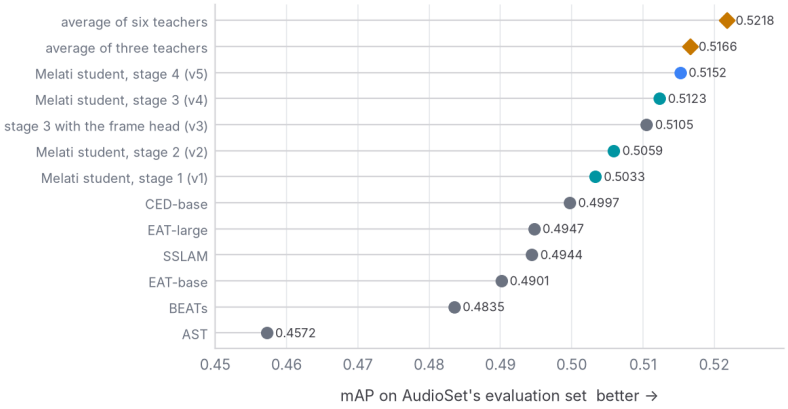


FIGURE 36 AudioSet taggers on the same 17,141 evaluation clips, in fp32: six released models, the two averages the student learned from (three teachers for stages 1 to 3, six for stage 4), the student after each stage of training, and stage 3 trained with the frame head beside it. In brackets, the version each student was released as.

The student is EAT-large, trained to its teachers' average probability for each class with binary cross-entropy: the three taggers above for three stages, and six for the fourth (below). AudioSet's own clip labels are not used, and no clip of the evaluation set is trained on. Four stages took it from EAT-large's own 0.4947 to 0.5152 (Figure 37):

1. 10 epochs on AudioSet's strongly labelled training clips (102,123, every event timed by hand) and its balanced set (11,397): 0.5033, already above every released model.
2. 3 epochs with 617,321 clips of the unbalanced set added (730,841 clips, 2,011 hours): 0.5059.
3. 5 epochs on the first set again, without mixup: 0.5123.
4. 3 epochs on the second set again, without mixup, toward six teachers: 0.5152.

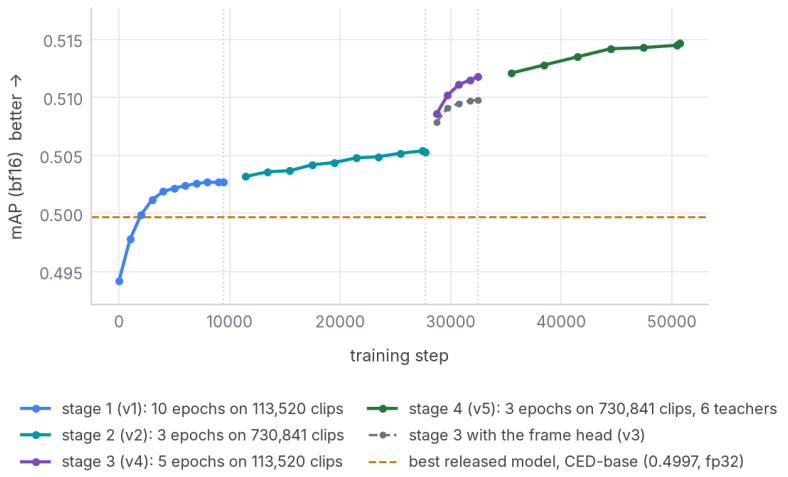


FIGURE 37 The student's mAP at each evaluation (bf16, the moving average of its weights), stage after stage. Grey dashes: stage 3 trained with the frame head beside it. Orange dashes: the best released model, in fp32. In brackets, the version each stage was released as.

Stage 3 was first trained with a second head beside the first. That head reads the encoder's

tokens every 160 ms and learns where each sound is from the hand-timed events. With it, the clip head reached 0.5105. Trained again with only that loss switched off, stage 3 reaches 0.5123. So the timed events did not lift the clip head. The lift over stage 2 comes from the rest of the stage: another learning-rate cycle on the first set, without mixup. The frame head also places sounds less well than PretrainedSED: on the 16,896 strongly labelled test clips its frame mAP is 0.4102, against 0.4261 for PretrainedSED's average, over the 364 classes both have. So stage 3 was released without the frame head.

A tag in a transcript also needs to know where its sound is. Read off 2 s windows, the stage 3 tagger scores 0.2970 frame mAP on those test clips, barely above the 0.2688 it scores by giving every 160 ms its probability for the whole clip. The mean of PretrainedSED's taggers and the frame head scores 0.4410, above either alone. Weighting it by the stage 3 tagger's probability for the clip lowers it (0.4208 for PretrainedSED). So the dataset's sound tags take which sounds a segment holds from the stage 3 tagger, and where they are from that mean (Section 7).

A larger student did not help. Dasheng-1.2B [Dinkel et al., 2024b] is a self-supervised encoder nearly four times EAT-large's size, with no AudioSet training of its own. Given a new head and the same 730,841 clips, it reached 0.4108 after 2.96 of its 6 epochs (step 18,000), and we stopped it there: its last 3,000 steps had added 0.0094. EAT-large starts from a model already trained on AudioSet (0.4947 on its own).

Better teachers did help. A student learns its teachers' average, and stage 3 ended 0.43 of a point under the three's, so we looked for teachers that raise it. Stage 3 and its frame-head twin, added to the three, lift the average to 0.5188. BEATs, the weakest of them alone, lifts the three to 0.5200; it was pre-trained differently from the rest. PretrainedSED's frame taggers, read at clip level (each class's highest frame), score 0.365 to 0.375 and lower every one of the 56 averages they join (the three's to 0.5130 at best). All six average 0.5218. Stage 4 trains stage 3 toward that average, reaching 0.5152: 0.29 of a point above stage 3 and still 0.66 of a point under its teachers. That is v5, the current release, under MIT as EAT is: `https://huggingface.co/Scicom-intl/Melati-AudioSet-EAT-Large`. The earlier stages were released there before it: stages 1 and 2 as v1 and v2, stage 3's frame-head twin as v3 and stage 3 as v4.

The six teachers' average is released as well, each model run with its own preprocessing: `https://huggingface.co/Scicom-intl/Melati-AudioSet-Ensemble`. Loaded back from the Hub it scores 0.5219 in fp32 and 0.5219 in bf16, for six forward passes and 1.2B parameters; the 0.5218 above averages probabilities stored in 16 bits. CED-base's feature extractor sets its decibel floor under the loudest clip it is given, so there each clip's floor is set under its own, as for a clip given alone, and CED-base alone reads 0.4993 (0.4997 in batches of 64). Left out one at a time, BEATs costs the ensemble most (0.0030), then CED-base (0.0025). EAT-large costs nothing (−0.0001): without it the other five average 0.5220. Stage 3 and its twin were both trained from it.

The members are rebuilt there in plain PyTorch, so loading the ensemble fetches only their weights and runs none of their repositories' code. An inference engine ships with it: every GPU holds all six members in bf16, and the 10 s windows of concurrent requests are batched together. Through it the evaluation set scores 0.5220 mAP, at 282 clips a second on 4 GPUs (2,820 times real time). One GPU tags 72 a second, about what the plain bf16 call does in batches of 64 (69): a clip costs 1.25 TFLOP, so the GPU's compute is the limit.

6 Against Emilia-Pipe on the same recordings

We ran upstream Emilia-Pipe, its code unchanged, on two sets: twelve VoxConverse recordings [Chung et al., 2020] (2.31 hours of debates, news and talk shows), whose reference speaker turns measure purity, and 200 Malaysian podcast recordings. Melati is read twice: everything it exports, and the subset upstream's filter would keep, because Melati filters by query rather than before export.

6.1 VOXCONVERSE

Upstream keeps 20% of the input; Melati exports 78%, and its Emilia-filtered subset 19% (Figure 38). Of the speech in which exactly one reference speaker talks, Melati keeps 87%, upstream 22%. Part of Melati's share is set by its transcriber, because export drops a segment whose transcript is empty: the released model leaves 4.3% of the 822 segments empty, the Indic + KL model alone 13.7%, most of them English speech it stopped on (Section 4).

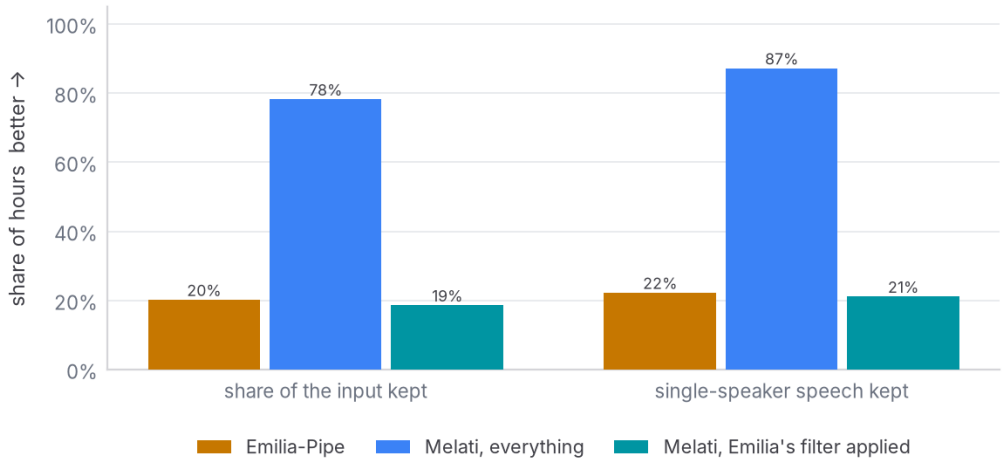


FIGURE 38 Hours kept, as a share of the input, and of the reference single-speaker speech.

Purity is measured against the reference turns (Figure 39): the share of a segment's reference speech in which only its main speaker talks, a second speaker present for half a second or more, and overlap of 0.2 s or more.

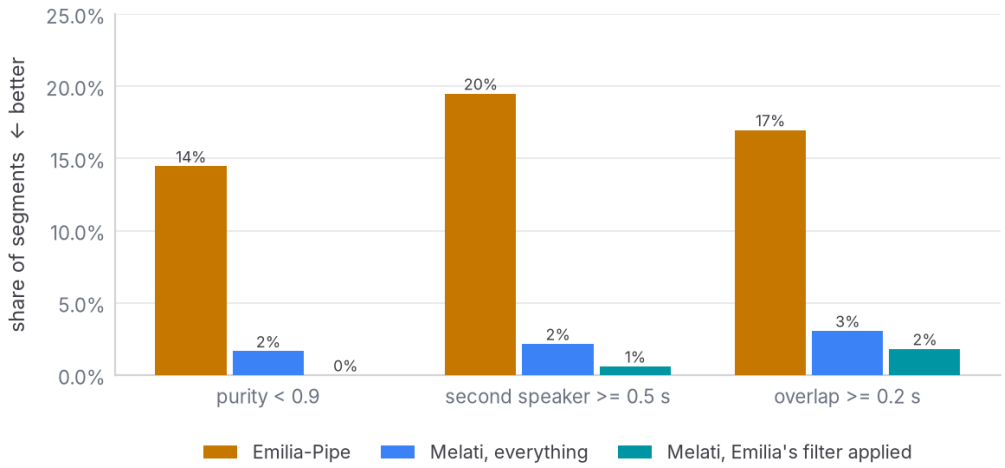


FIGURE 39 Segments that carry more than one voice, by three measures.

A segment that starts or ends inside a word pairs audio with text it does not contain. Each pipeline's own transcript, aligned to its own audio, shows where the first and last words fall (Figure 40): upstream starts mid-word in 37.1% of its segments, Melati in 9.2%.

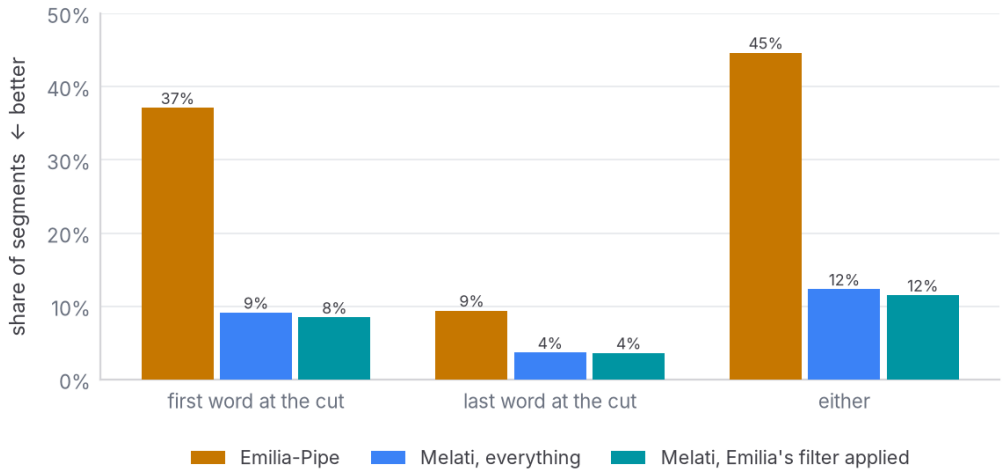


FIGURE 40 Segments whose first or last aligned word is within 20 ms of the cut.

Melati's segments are shorter, because it cuts in pauses and drops overlap instead of keeping whole turns (Figure 41); 28% of them are under three seconds, a length upstream's filter removes.

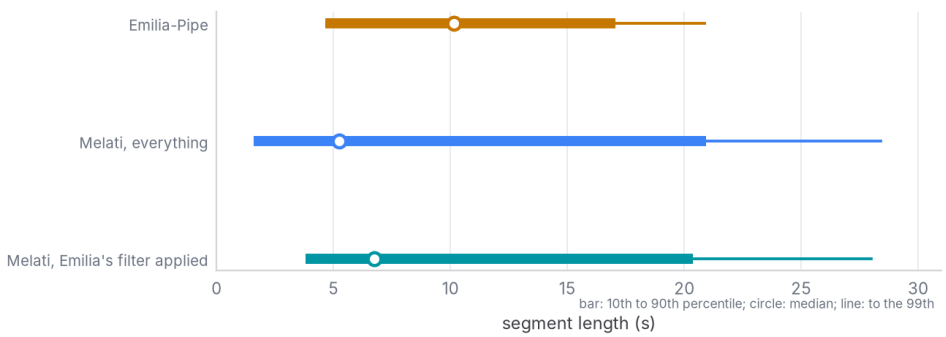


FIGURE 41 Segment length per pipeline.

6.2 MALAYSIAN PODCASTS

On the 200 podcasts (21.10 hours), upstream keeps 25% of the audio and Melati exports 74%; the part of Melati's export that upstream's own filter would pass is 19% (Figure 42).

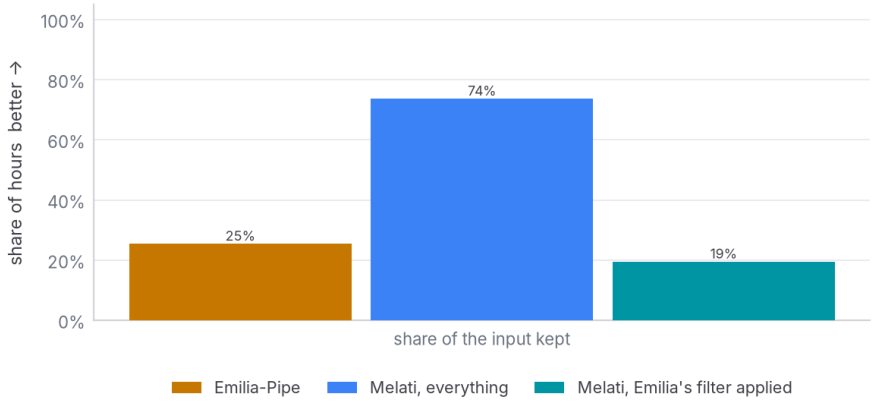


FIGURE 42 Hours kept as a share of the input, 200 Malaysian podcasts.

Upstream writes MP3 at 32 kbit/s, which cuts the spectrum at 8.4 kHz; Melati writes FLAC, whose spectrum ends where the sources' does, at 11.7 kHz. Upstream's level normalisation leaves 20.5% of its segments at full scale and 13.3% past it; Melati none.

Words at the cut (Figure 43): upstream starts inside a word in 31.3% of its segments and Melati in 16.7%; at the end the order flips, 10.6% against 15.7%. Podcast guests talk over each other far more than VoxConverse's speakers, and where an overlap is cut out the speaker it interrupts is often mid-word even after the back-off; upstream keeps the overlap, and with it the second voice.

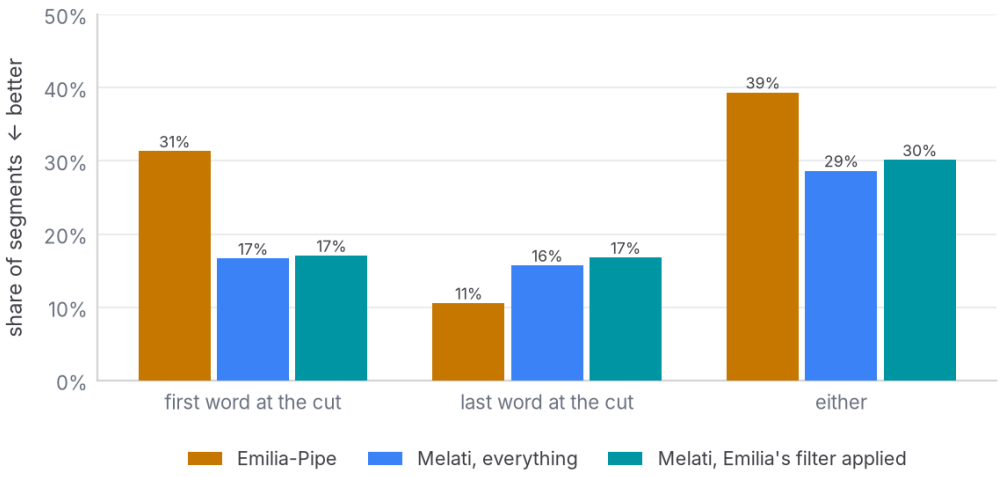


FIGURE 43 Segments whose first or last aligned word is within 20 ms of the cut, 200 Malaysian podcasts.

Of the laughs the event detector finds in the source recordings, 32% are still audible in Melati's export and 2% in upstream's, and Melati tags 31% of them. English words make up 8.9% of the words in Melati's Malay segments and 9.4% in upstream's: neither forces a language, so neither translates the code-switching away. Melati's segments are shorter (Figure 44), for the reasons they are on VoxConverse.

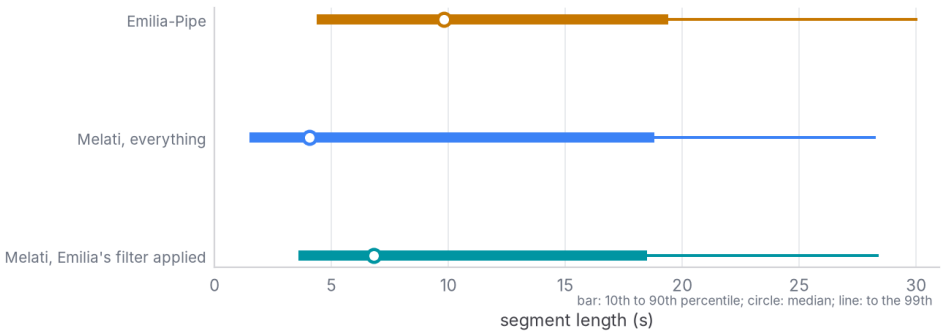


FIGURE 44 Segment length per pipeline, 200 Malaysian podcasts.

7 Melati v1

The run is under way: 7,640 of the source's 19,092 recordings (914 hours) have been through the pipeline so far, and the first 108 of Melati v1's 299 shards are published. Their 6,912 recordings (833 hours) hold 591 hours in 355,477 segments, 71% of the input, from 14,686 speakers counted per recording, or per ten-minute window in the 219 recordings (182 hours) diarized in windows. Malay is 95.6% of the hours and English 4.4%. Left out are the 8,010 segments (4.6 hours, 2.2% of all) whose transcript cannot be what was said: Chinese and Tamil that Whisper did not hear itself (6,939) or heard in a recording mostly in another language (972), which on these podcasts is every one of them; text mostly in a script its language is not written in (62, Quranic Arabic labelled Malay or English); and segments with no text (37). No score is filtered on: the core of Emilia-Pipe's filter (3 to 30 s, DNSMOS at least 3) would keep 181 hours of it, and a column counts the words aligned into under 25 ms, text the cut does not hold: two or more in 1.1% of the segments. It is published with the audio

inline, under CC BY-NC 4.0 as its source is, at https://huggingface.co/datasets/Scicom-intl/Melati.

Per hour of speech it tags 52.1 breaths, 29.5 laughs (29% of them over someone's words), 0.20 coughs, 0.90 sniffs and 0.09 throat clearings, marks 0.02 whispered spans, and writes 201 fillers (Figure 45). Emotion spans run from 98.6 happy ones an hour to 0.6 sad ones. It also writes the sounds around the speech that its AudioSet tagger hears (Section 5.4): 14.5 `[music]` tags an hour, and 290 others in all (beep 105, bleat 54, clapping 38, cricket 26, and 14 more kinds). Every segment also carries the tagger's labels as a column, and that column hears music in 22% of the segments. 77% of those are exported from the recording itself. The separator runs only where PANNs hears background, so these keep their music, mostly faint under the voice, and the column is how to leave them out.

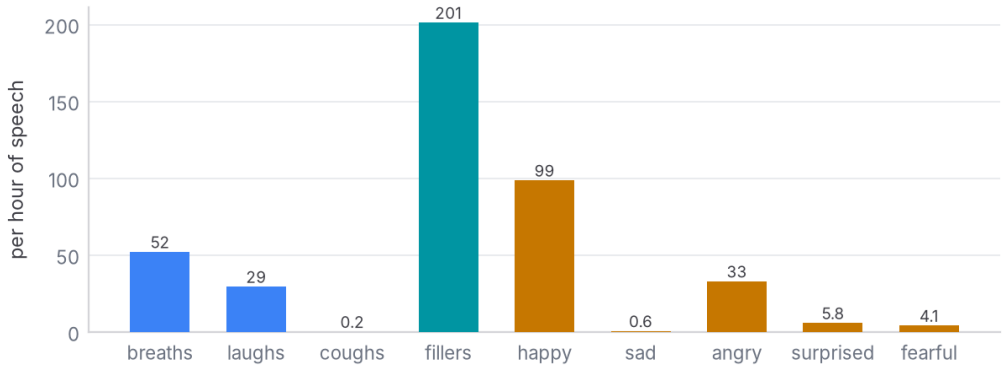


FIGURE 45 Melati v1's tags per hour of speech: non-verbal sounds (blue), fillers (teal), emotion spans (amber).

What a whole corpus taught the servers. So far the run has taken 19 hours on six shared GPUs, 48 times realtime overall (Figure 46). The pilot's 200 recordings never showed which server a whole corpus waits on, and each fix moved the wait to the next one:

1. The w2v-BERT emotion head ran one clip at a time and was the slowest model in the tag server by far. Clips of similar length batched, masked and pooled over each clip's own frames are the same arithmetic: on 400 real clips the features agree at cosine 0.9997 or better and the probabilities within 0.005, and it ran 3.0 times faster (WavLM 2.4).
2. Diarization reads a whole recording at once. Two long ones filled one server's KV cache while round-robin queued most of the waiting recordings behind them and its sibling had room. The client now sends each call to the replica with the fewest calls open, and diarization got two more replicas.
3. vLLM's default batch budget starts five Whisper encoder passes a step, and the filler stage transcribes every voiced gap alone, each a full 30-second pass. A larger budget and a third replica emptied its queue.
4. A diarization pass over 45 minutes holds 35 to 45% of a server's KV cache while it decodes at tens of tokens a second. Three filled a replica and nothing behind them was admitted, so these passes outlasted the client's half-hour timeout and failed five tries in a row, and giving them longer made it worse: a slow recording keeps its place in the client, and every recording in flight ended up waiting on diarization. A recording over 45

minutes is now ten-minute windows, and a shorter pass that gets no answer in two tries is too.

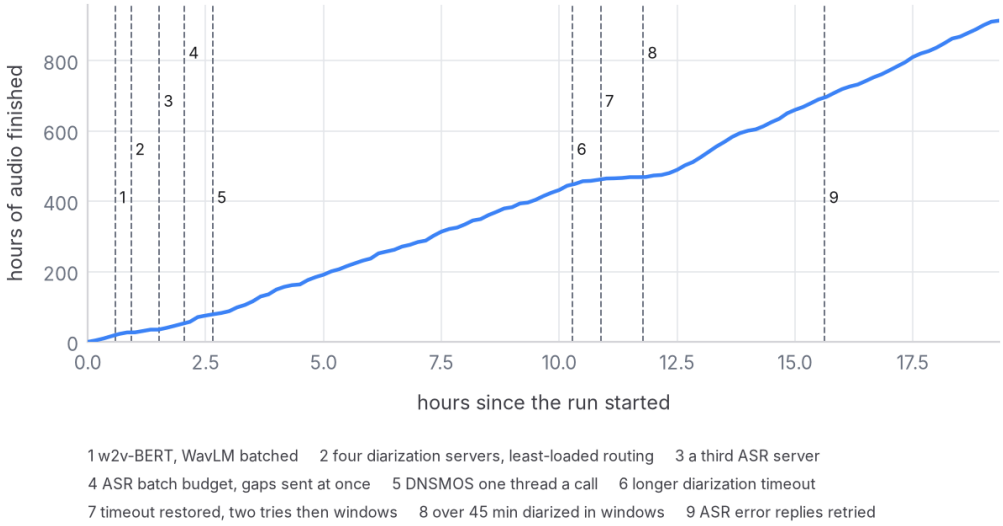


FIGURE 46 Hours of audio finished against hours of running, on six shared GPUs. Dashed: the changes made while it ran.

8 Limitations

No held-out emotion or non-verbal labels in Malay exist that we know of: every threshold was chosen on English and Polish speech, and the podcast rates behind the emotion thresholds are assumptions until the labelling set is labelled. Laughter two people share is overlap and is cut out. The released transcriber returns nothing for 0.7% of Malaysian clips and more of noisy speech, and still writes over 5.1% of real voice-free audio; the merge was chosen on the arms that separate the fine-tunes, which a held-out set would make firmer. Purity is measured on VoxConverse, which is not Malaysian. Throughput was measured on shared GPUs only. The fine-tune's training audio includes Malaysian podcast speech licensed for non-commercial use, and so is the checkpoint.

9 Conclusion

What a speech dataset keeps between the words decides what a generator trained on it can do. Melati keeps the laughs, fillers and emotion of podcast speech where it can show they are right, and keeps the audio whole. The transcriber it needed taught a second lesson: a model allowed to stop has to be measured where it is allowed to stop, or the benchmark grades the decoder's constraints instead of the model.

Acknowledgement

We thank Scicom (MSC) Berhad for the compute and the time to build this.

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